ExactFuse: Closed-Form Unfolding with Exact Operators for Sensor-Independent Hyperspectral and Multispectral Image Fusion
Supplementary Material

Anonymous CVPR submission

Equations, theorems, sections and tables of the main paper are referred to by their numbers there (*e.g.* Theorem 1, Tab. 1); items of this supplement carry the prefix S.

S1. The exact adjoint of reflect padding

Every gradient step on $\|\mathcal{D}\mathbf{X} - \mathbf{Y}_{\mathrm{H}}\|^2$ needs $\mathcal{D}^\top = \mathcal{P}^\top\mathcal{K}^\top\mathcal{S}^\top$, where $\mathcal{P}$ pads by p pixels, $\mathcal{K}$ is a valid correlation with the $(2p+1)\times(2p+1)$ kernel and $\mathcal{S}$ keeps every s-th pixel from index zero. $\mathcal{S}^\top$ inserts zeros and $\mathcal{K}^\top$ is a full convolution; together they are a strided transposed convolution with output padding $s-1$. The adjoint of reflect padding adds each padded margin back onto the pixels it was copied from. For a padded row index $j \in [0, H+2p)$ the source row is $p-j$ for $j<p$, $j-p$ in the interior and $2H-2+p-j$ in the bottom margin, hence

$$(\mathcal{P}^\top\mathbf{g})_i = \mathbf{g}_{i+p} + [1 \le i \le p]\,\mathbf{g}_{p-i} + [H-1-p \le i \le H-2]\,\mathbf{g}_{2H-2+p-i}, \tag{S1}$$

applied to rows and then to columns. With (S1), the operator in the network is exactly the simulation operator, and $\langle\mathcal{D}\mathbf{X}, \mathbf{Y}\rangle = \langle\mathbf{X}, \mathcal{D}^\top\mathbf{Y}\rangle$ holds to a relative error below 10^{-13} in double precision at every scale factor. $\mathcal{R}^\top$ is the transpose of the response matrix.

S2. Proofs

Theorem 1. (a) By (4), $\mathcal{D}_\theta^\top\mathcal{D}_\theta$ applies $\mathbf{D}_{\mathrm{r}}^\top\mathbf{D}_{\mathrm{r}}$ and $\mathbf{D}_{\mathrm{c}}^\top\mathbf{D}_{\mathrm{c}}$ to the two spatial modes and the identity to the spectral mode, and $\mathcal{R}_\phi^\top\mathcal{R}_\phi$ applies $\mathbf{r}_\phi^\top\mathbf{r}_\phi$ to the spectral mode alone. Each factor is diagonalized by the corresponding factor of $\mathcal{Q}$, and an identity factor by any orthonormal basis. (b) In this basis the matrix of (2) is diagonal with entries $1 + \alpha_k\lambda_i^{\mathrm{r}}\lambda_j^{\mathrm{c}} + \beta_k\nu_b > 0$. (c) $\mathcal{N}(\mathcal{D}_\theta) \cap \mathcal{N}(\mathcal{R}_\phi) = \mathcal{N}(\mathcal{D}_\theta^\top\mathcal{D}_\theta + \mathcal{R}_\phi^\top\mathcal{R}_\phi)$, whose eigenvalues $\lambda_i^{\mathrm{r}}\lambda_j^{\mathrm{c}} + \nu_b$ are sums of nonnegative terms and vanish only when both terms do. The basis tensors with $\lambda_i^{\mathrm{r}}\lambda_j^{\mathrm{c}} > 0$ span the row space of $\mathbf{D}_{\mathrm{r}} \otimes \mathbf{D}_{\mathrm{c}}$, of dimension hw, and those with $\nu_b > 0$ the row space of $\mathbf{r}_\phi$, of dimension M. The orthogonal projector onto the null space is therefore $(\mathbf{I} - \mathbf{P}_{\mathrm{s}}) \otimes (\mathbf{I} - \mathbf{P}_{\mathrm{r}} \otimes \mathbf{P}_{\mathrm{c}})$ in mode order, with $\mathbf{P}_{\mathrm{s}}, \mathbf{P}_{\mathrm{r}}, \mathbf{P}_{\mathrm{c}}$ the projectors onto the row spaces of $\mathbf{r}_\phi, \mathbf{D}_{\mathrm{r}}, \mathbf{D}_{\mathrm{c}}$, and its dimension is $(B-M)(HW-hw)$. □

Theorem 2. (a) $\tilde{\mathbf{A}}\mathbf{X}^* - \mathbf{Y} = \mathbf{0}$ makes both updates vanish. (b) With $\mathbf{B} = \mathbf{W}_k^{1/2}\tilde{\mathbf{A}}$, a direct computation gives $T(\mathbf{X}^*) - \mathbf{X}^* = (\mathbf{I} + \mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{e}$ with $\mathbf{e} = \mathbf{W}_k^{1/2}(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*$. In the singular value decomposition of $\mathbf{B}$, $(\mathbf{I}+\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$ has singular values $b_i/(1+b_i^2) \le \frac{1}{2}$. Similarly $G(\mathbf{X}^*) - \mathbf{X}^* = \mathbf{B}^\top\mathbf{e}$, whose norm is at most $\|\mathbf{B}\|\,\|\mathbf{e}\|$. (c) T_k is nonexpansive, since $\|T_k(\mathbf{V}) - T_k(\mathbf{V}')\| = \|(\mathbf{I} + \mathbf{B}^\top\mathbf{B})^{-1}(\mathbf{V} - \mathbf{V}')\| \le \|\mathbf{V} - \mathbf{V}'\|$. With $e_k = \|\mathbf{X}_k - \mathbf{X}^*\|$ and $(\mathcal{I} + \mathcal{P}_k)(\mathbf{X}^*) = \mathbf{X}^*$,

$$\begin{aligned} e_k &\le L_k\|T_k(\mathbf{X}_{k-1}) - \mathbf{X}^*\| \\ &\le L_k\big(\|T_k(\mathbf{X}_{k-1}) - T_k(\mathbf{X}^*)\| + \|T_k(\mathbf{X}^*) - \mathbf{X}^*\|\big) \le L_k\big(e_{k-1} + \tfrac{1}{2}\|\mathbf{e}_k\end{aligned}$$

and unrolling from $e_0 = 0$ gives the bound. The gradient step is not nonexpansive in general, so the bound has no counterpart for it. □

Corollary 1. A low-resolution pixel is a weighted sum of the $(2p+1)\times(2p+1)$ high-resolution pixels around its sampling position. Two border rules give the same value whenever this window lies inside the image, i.e., unless the sampling position is within p pixels of the border; with sampling positions $0, s, 2s, \ldots$ this concerns the first $\lceil p/s \rceil$ low-resolution rows and columns (and, at the far border, those whose window crosses it). □

Theorem 3. (a) On vectorized images, $\mathcal{D}_\theta = \mathbf{I}_B \otimes \bar{\mathcal{D}}_\theta$ and $\mathcal{R}_\phi = \mathbf{r}_\phi \otimes \mathbf{I}_N$, with $\bar{\mathcal{D}}_\theta$ the single-band spatial operator and $N = HW$. By the mixed-product rule, $(\mathbf{I}_M \otimes \bar{\mathcal{D}}_\theta)(\mathbf{r}_\phi \otimes \mathbf{I}_N) = \mathbf{r}_\phi \otimes \bar{\mathcal{D}}_\theta = (\mathbf{r}_\phi \otimes \mathbf{I}_{N/s^2})(\mathbf{I}_B \otimes \bar{\mathcal{D}}_\theta)$. With $\mathbf{Y}_{\mathrm{M}} = \mathcal{R}_{\phi^*}\mathbf{X}$ and $\mathbf{Y}_{\mathrm{H}} = \mathcal{D}_{\theta^*}\mathbf{X}$, $\mathcal{D}_{\theta^*}\mathbf{Y}_{\mathrm{M}} = \mathcal{R}_{\phi^*}\mathcal{D}_{\theta^*}\mathbf{X} = \mathcal{R}_{\phi^*}\mathbf{Y}_{\mathrm{H}}$. (b) By (a), $\mathbf{r}(\theta, \phi) = \mathcal{D}_\theta\mathcal{R}_{\phi^*}\mathbf{X} - \mathcal{R}_\phi\mathcal{D}_{\theta^*}\mathbf{X} = \mathcal{C}_{\theta\phi}\mathbf{X}$ with $\mathcal{C}_{\theta\phi} = \mathbf{r}_{\phi^*} \otimes \bar{\mathcal{D}}_\theta - \mathbf{r}_\phi \otimes \bar{\mathcal{D}}_{\theta^*}$, so $J = 0$ exactly on its null space. Suppose $\mathcal{C}_{\theta\phi} = \mathbf{0}$ and apply it to $\mathbf{X} = \mathbf{e}_b \otimes \mathbf{u}$, a single band b with spatial pattern $\mathbf{u}$: $(\mathbf{r}_{\phi^*})_{:b} \otimes \bar{\mathcal{D}}_\theta\mathbf{u} = (\mathbf{r}_\phi)_{:b} \otimes \bar{\mathcal{D}}_{\theta^*}\mathbf{u}$ for all $\mathbf{u}$. If $\bar{\mathcal{D}}_\theta \ne \bar{\mathcal{D}}_{\theta^*}$, then $\bar{\mathcal{D}}_\theta \ne c\,\bar{\mathcal{D}}_{\theta^*}$ for every scalar c: both kernels sum to one and reflect padding preserves constants, so both operators map a constant image to the same constant, which forces $c = 1$. Two linear

maps of rank greater than one that are not scalar multiples of each other map some $\mathbf{u}$ to linearly independent images, and for this $\mathbf{u}$ the identity forces $(\mathbf{r}_{\phi^*})_{:b} = (\mathbf{r}_{\phi})_{:b} = \mathbf{0}$, impossible for a response that sums to one. If $\bar{\mathcal{D}}_\theta = \bar{\mathcal{D}}_{\theta^*}$, the identity gives $(\mathbf{r}_{\phi^*})_{:b} = (\mathbf{r}_{\phi})_{:b}$ for every b. Hence $\mathcal{C}_{\theta\phi} \neq \mathbf{0}$ unless both operators coincide, and its null space is a proper subspace. Examples: a spatially constant image $\mathbf{X} = \mathbf{s} \otimes \mathbf{1}$ gives $\mathcal{C}_{\theta\phi^*}\mathbf{X} = \mathbf{r}_{\phi^*}\mathbf{s} \otimes (\bar{\mathcal{D}}_\theta\mathbf{1} - \bar{\mathcal{D}}_{\theta^*}\mathbf{1}) = \mathbf{0}$ for every θ; an image with equal bands $\mathbf{X} = \mathbf{1} \otimes \mathbf{u}$ gives $\mathbf{r}_\phi\mathbf{1} = \mathbf{1}$ for every ϕ, so $\mathcal{C}_{\theta^*\phi}\mathbf{X} = \mathbf{0}$. Over a finite set of injectively parametrized candidates the union of these null spaces still has measure zero, which recovers uniqueness on a grid for almost every $\mathbf{X}$. (c) Write $\vartheta = (\theta, \phi)$ and $F(\vartheta, \mathbf{N}) = \mathbf{G}(\vartheta, \mathbf{N})^\top \mathbf{r}(\vartheta, \mathbf{N})$, the gradient of $J/2$, where the noise enters as $\mathbf{r}(\vartheta, \mathbf{N}) = \mathbf{r}(\vartheta, \mathbf{0}) + \mathcal{D}_\theta \mathbf{N}_{\mathrm{M}} - \mathcal{R}_\phi \mathbf{N}_{\mathrm{H}}$. At $(\vartheta^*, \mathbf{0})$, $\mathbf{r} = \mathbf{0}$, so $\partial F/\partial\vartheta = \mathbf{G}^\top\mathbf{G} + \sum_i r_i \nabla^2 r_i = \mathbf{G}^\top\mathbf{G}$, which is invertible by assumption. The implicit function theorem gives a unique smooth stationary point $\hat{\vartheta}(\mathbf{N})$ near ϑ^* with $\hat{\vartheta} - \vartheta^* = -(\mathbf{G}^\top\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{n} + O(\|\mathbf{N}\|^2) = -\mathbf{G}^+\mathbf{n} + O(\|\mathbf{N}\|^2)$, and $\|\mathbf{G}^+\mathbf{n}\| \leq \|\mathbf{n}\|/\sigma_{\min}(\mathbf{G})$; under white noise its covariance is $\mathbf{G}^+\mathbf{\Sigma}_\mathbf{n}\mathbf{G}^{+\top}$. The zero is isolated because $\mathbf{r}(\vartheta, \mathbf{0}) = \mathbf{G}(\vartheta - \vartheta^*) + O(\|\vartheta - \vartheta^*\|^2)$ gives $\|\mathbf{r}\| \geq \sigma_{\min}(\mathbf{G})\|\vartheta - \vartheta^*\| - O(\|\vartheta - \vartheta^*\|^2) > 0$ for small $\vartheta \neq \vartheta^*$. The second-order term contains the bias of the next paragraph. □

Corollary 2. With $\tilde{\mathbf{A}} = \mathbf{A}_{\hat{\vartheta}}$, $(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^* = -\mathbf{A}'(\hat{\vartheta} - \vartheta^*) + O(\|\hat{\vartheta} - \vartheta^*\|^2)$, so $\|\mathbf{e}\| \leq \|\mathbf{W}_k^{1/2}\mathbf{A}'\|\,\|\hat{\vartheta} - \vartheta^*\|$ to first order; Theorem 2(b) and part (c) give the bound. □

Noise in the criterion J. With independent noise, $\mathbb{E}\,J(\theta, \phi) = \|\mathbf{E}_{\theta\phi}\mathbf{X}\|^2 + \frac{N}{s^2}\big(M\varsigma_{\mathrm{M}}^2\|\mathbf{k}_\theta\|^2 + \varsigma_{\mathrm{H}}^2\|\mathbf{r}_\phi\|_F^2\big)$ up to border terms, where $\mathbf{k}_\theta$ is the blur kernel. The noise term decreases with θ (a wider kernel has a smaller norm) and barely depends on ϕ, so noise biases $\hat{\theta}$ upward by an amount that vanishes as the SNR grows.

Proposition 1. Let $\varsigma_{bij} = \alpha_k \lambda_i^{\mathrm{r}} \lambda_j^{\mathrm{c}} + \beta_k \nu_b$, the eigenvalues of $\tilde{\mathbf{A}}^\top \mathbf{W}_k \tilde{\mathbf{A}}$, and e^f the numerator of Γ, so $\frac{1}{2} < e^f < 2$. (a) $\mathbf{G}_k = \mathbf{0}$ at $\mathbf{X}^*$. (b) The step maps a difference $\mathbf{V} - \mathbf{V}'$ to $\mathcal{Q}\big((1 - \Gamma\varsigma) \odot \mathcal{Q}^{-1}(\mathbf{V} - \mathbf{V}')\big)$, and $0 \leq \Gamma\varsigma = e^f\varsigma/(1 + \varsigma) < 2$, so every factor lies in $(-1, 1]$. (c) The displacement is $\mathcal{Q}\big(\Gamma \odot \mathcal{Q}^{-1}(\mathbf{B}^\top \mathbf{e})\big)$, whose operator norm is $\max e^f\sqrt{\varsigma}/(1 + \varsigma) \leq 2 \cdot \frac{1}{2} = 1$. □

S3. Data and protocol

Chikusei. The scene [24] has 2517×2335 pixels and 128 bands from 363 to 1018 nm. The split sizes follow [5]: eight 256×256 test images, 64 validation patches and 832 training patches of 64×64. Their locations are not specified there, so we fix them: we take the central 2048×2048 crop, which has no no-data border, and divide it by its global maximum (15,133 digital numbers, DN). The test images are the eight 256×256 tiles of the top 256 rows, the validation patches tile rows 272–400, and training draws random 64×64 crops with dihedral augmentation from rows 416–2048. Sixteen-row gaps keep the regions disjoint.

Pavia Centre. The ROSIS scene [10] has 1096×715 pixels and 102 bands between 430 and 860 nm. As in [5] we keep bands 11–102 (92 bands) and use a four-band multispectral image. The benchmark files distributed with PSRT [6] were not accessible, so we rebuild the benchmark from the original scene to the same specification: division by the maximum of the retained bands (8,000 DN), four 256×256 test tiles in the top-left 512×512 region, 63 validation patches on a 32-pixel grid in rows 528–656, and random training crops from rows 672–1096 and from the strip to the right of the test region (rows 0–512, columns 528–715), with 16-pixel gaps.

CAVE and Harvard. CAVE [23] has 32 indoor scenes of 512×512 pixels and 31 bands (400–700 nm); the first 20 scenes train and the last 12 test, as in MHF-Net, MoG-DCN and DCT. Harvard [4] has 1040×1392 images with 31 bands (420–720 nm); 30 images train and the central 512×512 crops of 20 images test, each image divided by its own maximum. Validation patches come from the bottom-right corners of the training images. Both are paired with a Nikon RGB response.

NEON (real data). The NEON airborne observation platform [12] records an imaging spectrometer (1 m) and an RGB camera (0.1 m) on the same flights. We use eight tiles of the SERC site (2019), 120 spectrometer bands between 400 and 1000 nm, and the camera image resampled to 0.25 m. The camera's spectral response is not published, so we fit a unit-sum nonnegative response of the three camera channels to the spectrometer bands on co-located 4 m blocks (R^2 between 0.45 and 0.79). At reduced resolution, the measured 1 m spectrometer image is the reference, the LR-HSI is simulated from it at $\times 4$, and the measured camera image is the HR-MSI; at full resolution the measured 1 m image is fused with the 0.25 m camera image and evaluated without a reference (D_λ, D_s, HQNR [2]). Table S18 lists the results.

A second site. To test across scenes, the networks trained on SERC (Maryland) are tested without retraining on SJER (California, 2019), a savanna site with other vegetation, soil and illumination. The camera response is fitted on three SJER tiles (R^2 0.69–0.78), and 20 crops of five other tiles form the test set; on two of them the fitted response explains

the camera poorly ($R^2 \leq 0.6$), and we keep them. ExactFuse reaches 36.91 dB, 1.20 dB above the best of the 4 retrained networks tested there (DHIF-Net, 35.71 dB), and has the lowest SAM (Tab. S19).

Spectral responses. WorldView-2 (Chikusei) and IKONOS (Pavia Centre) publish their band responses only as plots with band edges. We model each band as a flat top with logistic flanks (4 nm roll-off) that pass through the published edges, at 5% response for WorldView-2 and at 50% for the IKONOS band limits (blue 445–516, green 506–595, red 632–698 and near infrared 757–853 nm), and normalize every band to unit sum.

Degradation. Every image, including every training crop, is degraded on its own with (1): reflect padding, a normalized Gaussian blur (7×7, $\sigma = 2$ at $\times 4$; $(2s-1) \times (2s-1)$, $\sigma = s/2$ at $\times s$), decimation from index zero, and the spectral response. At $\times 16$ all methods train on 128×128 patches with batch 4.

S4. Baselines

All published networks are built from their authors' repositories, cloned at run time; only the numbers of input and output bands and the scale factor are changed. PSRT: github.com/Deng-shangqi/PSRT (9c2be90); DCT: github.com/qingma2016/DCTransformer (bd75b15); MIMFormer: github.com/meiruni/MIMFormer (0904158); DHIF-Net: github.com/TaoHuang95/DHIF-Net (6434366), whose code covers $\times 8$ and $\times 16$, so at $\times 4$ we use one of the two stride-4 layers of its $\times 16$ operator; SMGU-Net: github.com/yansql/SMGU-Net (115ce77), written for $\times 4$ only; CLSNet: github.com/HengYang01/CLSNet (6b13371), whose upsampling layer is set to the scale factor; OTPNet: github.com/tsharry2201/OTPNet (7c17964), written for pansharpening, with the HR-MSI in place of the panchromatic image; U2Net: github.com/PSRben/U2Net (32e214c), also written for pansharpening, with its input, upsampling and output layers resized; its attention spans all pixels, so a 256×256 test image does not fit a 16 GB T4, and it is tested in overlapping 128×128 tiles (32-pixel overlap, averaged); MIMO-SST: github.com/Freelancefangjian/MIMO-SST (06aacc1), whose final full-resolution output is used. DCT and MIMFormer use gradient checkpointing to fit a 16 GB T4, which does not change what they compute. SSR-NET is our re-implementation of [28].

Under the common recipe every baseline uses its published learning rate and an L1 loss with AdamW ($\beta = (0.9, 0.99)$, no weight decay, i.e., Adam), batch 8, a 1000-iteration warm-up and cosine decay over the budget, and no gradient clipping; ExactFuse uses the same pipeline with its own learning rate, weight decay and clipping. The authors' own recipes (Tab. S6) are read from their code: PSRT uses AdamW (learning rate 10^{-4}, decay 10^{-4}), clipping at 10, batch 32 and a constant learning rate; MIMFormer uses Adam (10^{-4}), batch 16 and step decay at 30, 60 and 90 of 101 epochs, mapped onto the time budget.

S5. ExactFuse details

Each prior (Fig. S1) is a three-level U-Net with widths 48, 96 and 192, two, three and four transformer blocks on the encoder levels and the bottleneck, mirrored decoder levels and two refinement blocks. A block (Fig. S2) follows Restormer [27]: transposed attention (Fig. S3) with 1, 2 and 4 heads on the three levels, followed by a gated depth-wise feed-forward network with expansion 2.66. ExactFuse has 10.5 M parameters for Chikusei; compared with the base model, the wider input convolution of each prior ($4B$ instead of $2B + M$ channels) adds about 0.3 M, and the gain networks of the data steps are negligible. Training uses AdamW ($\beta = (0.9, 0.99)$, weight decay 10^{-4}, peak learning rate 3×10^{-4}), a 1000-iteration warm-up, cosine decay to 10^{-3} of the peak at the end of the budget, gradient-norm clipping at 0.1, and an exponential moving average of the weights (decay 0.999). The image path, including all operator applications, runs in 32-bit precision and the priors in 16-bit mixed precision. The weights with the best validation PSNR are tested once, without self-ensemble.

S6. Gaps, evidence and metrics

Table S2 lists the gaps in current fusion networks that this work addresses, what ExactFuse does about each, and where the evidence is; Tab. S3 defines every metric.

S7. Additional results

Confidence intervals and per-image results. The 95% paired bootstrap interval of ExactFuse's mean per-image gain over the strongest retrained network excludes zero in every setting (Tab. S11); the smallest gain, 0.21 dB on Harvard, has the interval [0.11, 0.33] dB, and pooled over all test images the gain is 1.09 dB [0.88, 1.31]. ExactFuse has the lower SAM and ERGAS on every test image of every setting and the higher PSNR on all but one Harvard image (Tab. S12 lists Chikusei and Pavia Centre).

Training curves. Is the budget long enough? Figure S5 shows the validation PSNR of every network. The four strongest baselines gain at most 0.11 dB on Chikusei and 0.08 dB on Pavia Centre in the last quarter of their budget (MIMO-SST 0.02 dB, ExactFuse 0.03 dB on Chikusei), so they are close to converged. DCT and DHIF-Net, the slowest per iteration, still gain 1.37 and 1.13 dB and are under-

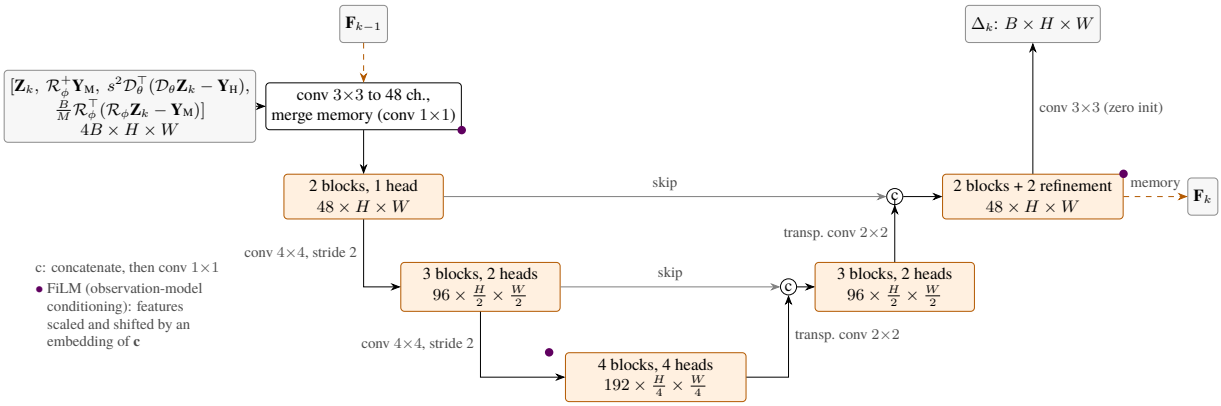


Figure S1. The prior $\mathcal{P}_k$ of stage k (widths for Chikusei, $B = 128$): a three-level U-Net of transformer blocks (Fig. S2). Its four inputs all lie in the B hyperspectral bands: the data-step output, the minimum-norm spectra $\mathcal{R}_\phi^+\mathbf{Y}_\mathrm{M}$ that reproduce the HR-MSI, and the residuals of both observations mapped back by the adjoints, so the prior does not depend on the multispectral sensor. The encoder halves the resolution twice with strided convolutions, the decoder restores it with transposed convolutions and merges the encoder features through skip connections; the previous stage's decoder features enter after the input convolution, and the last convolution starts at zero so that training starts from the HQS iteration.

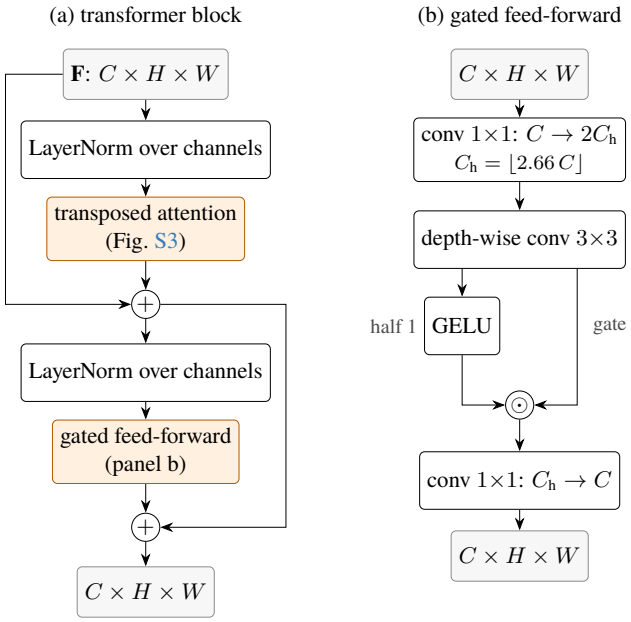


Figure S2. (a) A transformer block of the prior: layer normalization over channels, transposed attention and a gated feed-forward network, each in a residual branch. (b) The gated feed-forward network: the expanded features are split in two halves, and the GELU of one half gates the other.

trained, but end more than 3 dB below the strongest networks; the 8 h runs of the main paper test the effect of a longer budget.

Spectral regions without HR-MSI coverage. The WorldView-2 HR-MSI does not respond to 13 of the 128 Chikusei bands (Fig. S6); there only the LR-HSI and the prior inform the result. ExactFuse's mean RMSE in these bands is 20.9 DN, against 25.2 DN for MIMFormer, and 12.7 against 15.6 DN in the covered bands.

Blur of any shape. The identifiability residual $\mathcal{D}_k\mathbf{Y}_\mathrm{M} - \mathcal{R}_\phi\mathbf{Y}_\mathrm{H}$ is linear in the kernel k, so a kernel of any shape is the least-squares solution over all bands and low-resolution pixels, with taps that sum to one (5 ms per image). On noise-free pairs its relative error against the true kernel is 0.00% on Chikusei and 0.79% on Pavia Centre over the out-of-family blurs, and the network given it reaches the PSNR it reaches with the true kernel (any-shape row of Tab. 2). Its price is noise sensitivity: at 30 dB SNR the kernel error grows to 4.6% and 2.8%, and the PSNR is 0.10 and 0.16 dB below that with the isotropic estimate; at the training operator the two agree within 0.05 dB. A non-separable kernel loses the closed form, so the network takes its conjugate-gradient data step with the estimated kernel.

Misregistration. Table S15 compares the strategies on six conditions. The bilinear shift of the simulation also blurs the HR-MSI, which caps every registered result (ExactFuse 49.74 dB on Chikusei); with a sharp shift, as a sensor records it, registration brings ExactFuse to 54.56 dB, close to the 58.05 dB without misregistration. Noise is where the network trained on noise-free data falls behind MIMFormer, as in the robustness table; ExactFuse trained over the wider family, with noise and grid offsets, has the best registered result in every noisy condition but one, where it ties MIMFormer (36.90 against 36.92 dB). Putting the offset into ExactFuse's operator and shifting the result back is worse

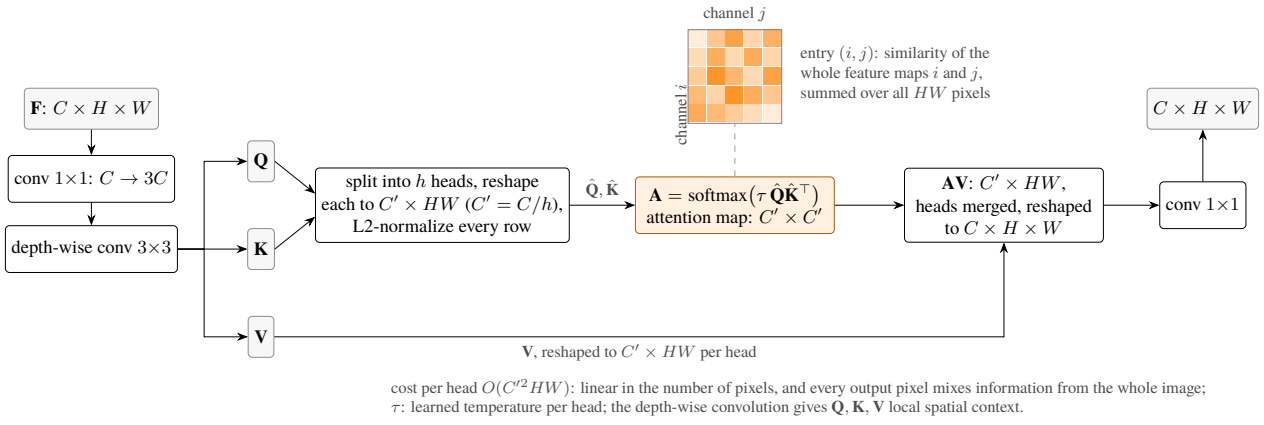


Figure S3. Multi-Dconv-head transposed attention [27], the token mixer of the prior. Queries, keys and values are computed by point-wise and depth-wise convolutions, and the attention map is formed between the C' feature channels of each head rather than between the HW pixels, so its size does not depend on the image size. Replacing it by a depth-wise convolution changes the PSNR by less than 0.1 dB in the ablation.

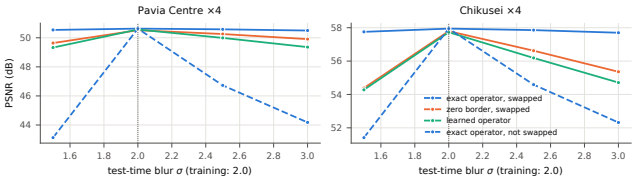


Figure S4. Test PSNR when the test blur differs from the training blur ($\sigma_0 = 2$, dotted): exact operator with and without the true blur, zero border with the true blur, and a learned operator, which cannot be replaced.

than registering the HR-MSI first (40.60 dB on Chikusei). The MIMFormer of this comparison is trained on noise-free data; registration-aware networks were not retrained.

Numerical check of the null-space split. On the test images, on random images and on bicubic errors, the round trip through the eigenbasis, the idempotence and symmetry of the projector, and the sum of the observed and null energies deviate from exactness by at most 6e-7 (single precision); the null-space component passes through the actual $\mathcal{D}$ and $\mathcal{R}$, reflect border included, with a relative norm of at most 7e-8; and the null-space fraction of the unknowns is 87.89% on Chikusei and 89.67% on Pavia Centre, as Theorem 1(c) predicts.

Corrupted observations. An exact data step trusts its observations, so corrupted pixels pass into the reconstruction. With local scene changes in the HR-MSI, dead LR-HSI columns and 35 dB noise, ExactFuse trained over the operator family reaches 37.23 dB on Chikusei; trained on corrupted pairs with a confidence-weighted data step, which replaces each observation by $w\mathbf{Y} + (1-w)\mathbf{AX}$ with a learned per-pixel weight w, it reaches 47.74 dB (GSA 34.41 dB). The gain comes from training on corrupted pairs: the same network without the confidence step reaches 47.76 dB. What the confidence step adds is a map of the corrupted pixels. The map $1 - w$ is a detector of the corrupted pixels: its ROC AUC against the corruption masks is 1.00 for the dead columns and 1.00 for the changed regions on Chikusei. It fails for changes in the four-band HR-MSI of Pavia Centre (AUROC 0.32), where the network still tolerates them (45.66 dB against 35.74 dB without confidence weighting) but does not localize them; dead columns are found there as well (1.00). A detector without learning localizes them. For consistent observations $\mathcal{D}\mathbf{Y}_\mathrm{M} = \mathcal{R}\mathbf{Y}_\mathrm{H}$ (Theorem 3(a)), so the energy of $\mathcal{D}\mathbf{Y}_\mathrm{M} - \mathcal{R}\mathbf{Y}_\mathrm{H}$ over the bands, with the estimated operators and normalized by its median, marks the low-resolution pixels where the two observations disagree. Its AUROC is 0.996 for the changed regions of Pavia Centre (0.980 with dead columns and 35 dB noise) and 0.965 for locally misregistered regions, where the confidence map reaches 0.47; on Chikusei it is 0.997 and 0.985. The residual localizes at the resolution of the LR-HSI and cannot tell which of the two observations is wrong, so it complements the learned confidence map.

Stability of the operator estimate. Table S10 checks Theorem 3(c) on the test images. The Jacobian has full rank for all five parameters, with condition numbers of 1.28 (Chikusei) and 1.47 (Pavia Centre) after scaling its columns, so the parameters are nearly uncorrelated. For the grid offsets and the response shift the first-order prediction matches the measured error; for the blur widths the measured error exceeds it as the SNR falls, because noise also enters the Jacobian and biases the blur estimate upward (next paragraph), an effect of second order that the first-order bound does not

Table S1. Blocks and sub-blocks of ExactFuse, in the order they run (shapes for Chikusei: $B = 128$ hyperspectral bands, $M = 8$ multispectral bands, scale $s = 4$, an $H \times W$ image; $h = H/s$, $w = W/s$).

Block	Sub-blocks	Input → output	Function
Observation model, set up once per image (no learned parameters)			
Operator estimate	grid search of Eq. (7) over the row and column blur widths, the sub-pixel grid offsets and the spectral-response shift	$(\mathbf{Y}_\mathrm{H}, \mathbf{Y}_\mathrm{M}) \to (\hat{\theta}, \hat{\phi})$	finds the operators under which $\mathcal{D}_\theta \mathbf{Y}_\mathrm{M}$ and $\mathcal{R}_\phi \mathbf{Y}_\mathrm{H}$ agree (Thm. 3); given operators skip this step
Exact operators	separable Gaussian blur with the simulation's reflect border, $\times s$ decimation, the spectral-response matrix, and their exact adjoints	$(\hat{\theta}, \hat{\phi}) \to \mathcal{D}_\theta, \mathcal{R}_\phi$	reproduce the simulation exactly, so the ground truth is a fixed point of every data step (Thm. 2)
Eigenbasis	eigendecompositions of $\mathbf{D}_\mathrm{r}^\top \mathbf{D}_\mathrm{r}$ $(H \times H)$, $\mathbf{D}_\mathrm{c}^\top \mathbf{D}_\mathrm{c}$ $(W \times W)$ and $\mathbf{r}_\phi^\top \mathbf{r}_\phi$ $(B \times B)$, cached per size	$\to \mathcal{Q}$, $\lambda^\mathrm{r}, \lambda^\mathrm{c}, \nu$	closed-form data step and explicit null space (Thm. 1)
Observation code	noise level of each observation (median absolute deviation of second differences), blur and shift	$\to \mathbf{c} \in \mathbb{R}^4$ ($\mathbb{R}^7$ with offsets)	conditions step sizes, gains and prior (FiLM) on the observation model
Unfolded network: $\mathbf{X}_0 =$ *bicubic upsampling of* $\mathbf{Y}_\mathrm{H}$*, then* $K = 3$ *stages*			
Data step	residuals of both observations, $\mathbf{G}_k = \alpha_k \mathcal{D}^\top(\mathbf{Y}_\mathrm{H} - \mathcal{D}\mathbf{X}_{k-1}) + \beta_k \mathcal{R}^\top(\mathbf{Y}_\mathrm{M} - \mathcal{R}\mathbf{X}_{k-1})$; three mode products into the eigenbasis; each component scaled by the gain $\Gamma_{k,bij}$ of Eq. (9); three mode products back	$\mathbf{X}_{k-1} \to \mathbf{Z}_k, B \times H \times W$	makes the estimate agree with both observations; keeps $\mathbf{X}^*$ fixed and is nonexpansive for any learned gains (Prop. 1); learns α_k, β_k and two two-layer perceptrons of the log-eigenvalues
Prior input	$[\mathbf{Z}_k,\ \mathcal{R}_\phi^+ \mathbf{Y}_\mathrm{M},\ s^2 \mathcal{D}^\top(\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_\mathrm{H}),\ \frac{B}{M}\mathcal{R}^\top(\mathcal{R}\mathbf{Z}_k - \mathbf{Y}_\mathrm{M})]$	$\to 4B \times H \times W$	every input in the hyperspectral bands: the prior does not depend on the multispectral sensor
Prior: stem	3×3 convolution to 48 channels; previous stage's decoder features merged by a 1×1 convolution; FiLM by $\mathbf{c}$	$\to 48 \times H \times W$	entry of the U-Net; cross-stage memory
Prior: encoder	2 transformer blocks (1 head); 4×4 convolution, stride 2; 3 blocks (2 heads); 4×4 convolution, stride 2	$\to 192 \times \frac{H}{4} \times \frac{W}{4}$	features at three scales
Prior: bottleneck	4 transformer blocks (4 heads); FiLM by $\mathbf{c}$	$192 \times \frac{H}{4} \times \frac{W}{4}$	context over the coarsest scale
Prior: decoder	2×2 transposed convolution, skip concatenation and 1×1 convolution, 3 blocks; again to full size, 2 blocks and 2 refinement blocks; FiLM	$\to 48 \times H \times W$, memory $\mathbf{F}_k$	restores the resolution with the encoder's detail
Prior: head	3×3 convolution, initialized to zero	$\to \Delta_k, B \times H \times W$	training starts from the data steps alone
Transformer block	layer norm; transposed attention (queries, keys and values by 1×1 and depth-wise 3×3 convolutions, a $C \times C$ attention map per head, learned temperature); layer norm; gated depth-wise feed-forward network (expansion 2.66); both in residual branches	$C \times H' \times W' \to$ same	mixes channels globally at a cost linear in the number of pixels (Figs. S2 and S3)
Update	$\mathbf{X}_k = \mathbf{Z}_k + \Delta_k$	$\to \mathbf{X}_k$	the prior adds what the data step cannot recover
Output	$\hat{\mathbf{X}} = \mathbf{X}_3$	$B \times H \times W$	the fused image, consistent with both observations

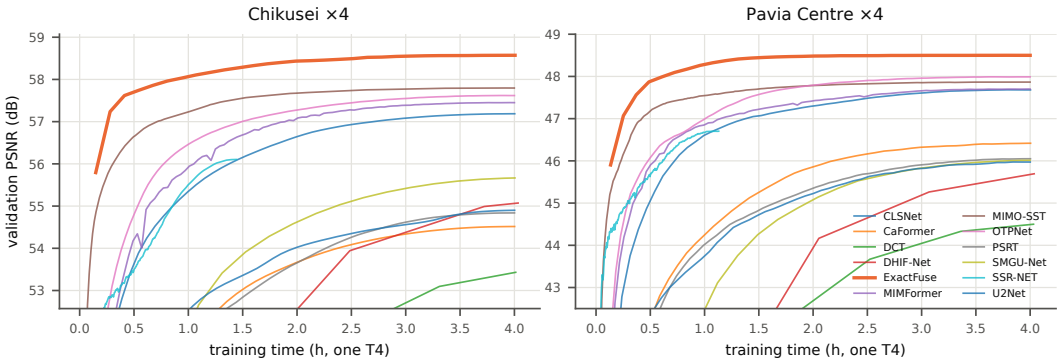


Figure S5. Validation PSNR of every network during its training budget (4 h on one T4; SSR-NET 1.5 h), Chikusei and Pavia Centre ×4.

Table S2. Gaps in deep hyperspectral and multispectral fusion, what ExactFuse does about them, and the evidence.

Gap	ExactFuse	Evidence
Blur, decimation and spectral response are learned or use zero padding, so they differ from the operators that made the data	exact operators with the simulation's border rule, a closed-form data step in their eigenbasis	Thm. 2; ablation rows "zero border" and "learned operator" (Tab. 4); test-time operator swap (Fig. S4)
Fused images do not reproduce their own inputs	every stage solves the data step exactly; the error is split into an observed and a null-space part	Obs. column of Tab. 1; agreement with the LR-HSI in Tab. 4
One network per sensor pair	every input of the prior lies in the hyperspectral bands; training over random spectral responses	sensor results (Sec. S7)
A fixed blur, response and noise level	training over a family of observation models, operators estimated from the pair, observation-model conditioning	Tab. 2
Inputs assumed co-registered	sub-pixel grid offsets in the operator family, estimated from the pair	offset and misregistration columns of Tab. 2
Why fuse at all, rather than super-resolve the LR-HSI	information budget of the HR-MSI; the same network with one input removed	remark after Thm. 1; bottom rows of Tab. 4
Comparisons copied across protocols and implementations	eleven networks retrained from their authors' code under one protocol and budget; recipe swap; three seeds; pooled signed-rank test; four datasets, three scale factors	Tabs. 1, 3 and Tab. S6

Table S3. Metrics. $\mathbf{X}$: reference, $\hat{\mathbf{X}}$: fused image, both scaled to $[0, 1]$ by the dataset's maximum; means over the test images.

Metric	Definition	Better
PSNR	$10 \log_{10}(1/\mathrm{MSE})$ over the whole cube, *i.e.* peak equal to the dataset's maximum	higher
SSIM	structural similarity per band, Gaussian window 11×11, $\sigma = 1.5$, data range equal to the image's own maximum (Sec. S7 also gives the PSRT definition with data range 1)	higher
SAM	mean angle between the reference and fused spectra of every pixel, in degrees	lower
ERGAS	$\frac{100}{s}\sqrt{\frac{1}{B}\sum_b (\mathrm{RMSE}_b/\mu_b)^2}$ with the band means μ_b	lower
Q2n	hypercomplex quality index [19] on 32×32 blocks with shift 32, computed on the digital numbers	higher
CC, SCC	correlation coefficient of the images, and of their Sobel gradient magnitudes	higher
RMSE	root-mean-square error in digital numbers, independent of the PSNR peak	lower
Cons.	$\mathrm{PSNR}(\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{H})$: how well the fused image reproduces the LR-HSI	higher
Obs.	mean squared error of the part of $\hat{\mathbf{X}} - \mathbf{X}$ in $\mathrm{row}(\mathcal{D}) + \mathrm{row}(\mathcal{R})$, which contradicts the observations (Thm. 1(c))	lower

Table S4. Properties of the closest methods (✓: yes, ○: partly or in another form, –: no; for ExactFuse ○ marks an optional head reported separately). SR: single-image super-resolution.

Method	Exact op.	True border	Closed form	Op. estimated	Any sensor	Misreg.	Uncertainty	Corrupt. inputs
DHIF-Net (TCI'22)	–	–	–	–	–	–	–	–
SMGU-Net (PR'25)	–	–	–	–	–	–	–	–
CaFormer (PR'26)	○	–	–	–	–	–	–	–
IR&ArF (TIP'25)	–	–	–	–	–	✓	–	–
USRNet (CVPR'20, SR)	✓	–	○	–	–	–	–	–
SSA (ICML'26)	–	–	–	–	○	–	–	–
SC-Net (arXiv'26)	✓	–	–	–	–	–	–	–
ExactFuse	✓	✓	✓	✓	✓	✓	○	○

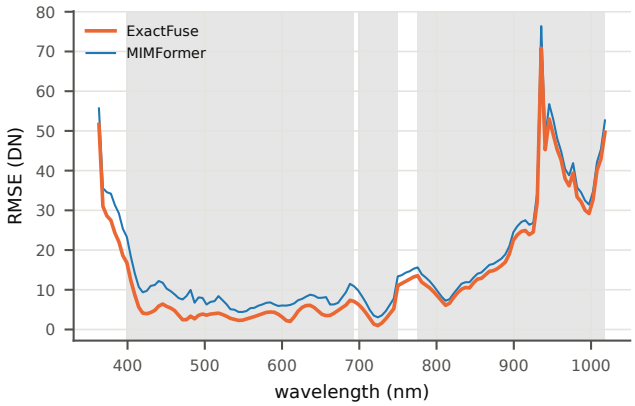


Figure S6. RMSE (DN) of every band on the Chikusei test images. Shaded: wavelengths to which the WorldView-2 HR-MSI responds.

contain. Even at 20 dB SNR the errors stay far below the grid step of 0.05 px. The conditioning is as good for two different blur widths (1.23) and with grid offsets and a response shift (1.29). For scenes in the null space of $\mathcal{C}_{\theta\phi}$ the corresponding columns of $\mathbf{G}$ vanish, as part (b) states: for spatially constant scenes the blur and offset columns fall to 3.5e-06 and 1.4e-13 (Chikusei), and for scenes with equal bands the response-shift column falls to 9.8e-07. The first-order relative operator residual used in the corollary matches the true one closely for blur errors up to 10% (1.16% against 1.07% at +10% on Chikusei).

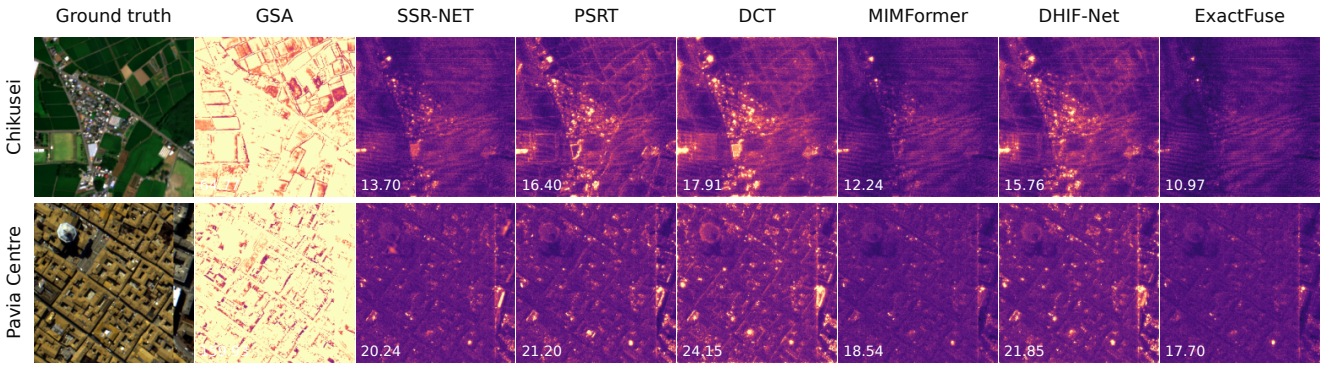


Figure S7. Mean absolute error over all bands (DN) on the test image of each dataset where ExactFuse has its lowest PSNR; color scale set by the retrained networks (99.5th percentile), so GSA saturates.

Table S5. Every ablation run (4 h each): ExactFuse with one change per row, one input removed, and the base model with its own steps towards ExactFuse and its operator ablation. Cons.: PSNR($\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{H}$). Swap: worst PSNR over test blurs $\sigma \in \{0.75, 1.25, 1.5\}\sigma_0$ when the true blur is given to the data steps (n/a: no operator to replace; –: not evaluated).

	Chikusei			Pavia Centre		
Variant	PSNR	Cons.	Swap	PSNR	Cons.	Swap
ExactFuse	58.05	93.4	57.93	50.63	82.5	50.57
gradient step instead of closed form	58.03	86.7	57.84	50.63	79.5	50.56
no data steps	57.42	67.5	n/a	50.30	68.1	n/a
+ null-space gate	57.97	96.9	57.82	50.58	87.0	50.48
+ gate, EON prior instead of U-Net	57.83	96.5	57.74	50.37	84.9	50.32
+ gate, transfer branch	58.07	96.2	57.93	50.55	85.0	50.39
base model (gradient step, $\mathbf{Y}_\mathrm{M}$ input)	57.95	83.9	57.70	50.63	77.6	50.50
+ closed-form step, learned gains	58.00	89.4	57.83	50.63	82.9	50.50
+ HR-MSI residual input	58.04	86.5	57.85	50.64	79.0	50.58
without data steps	57.65	73.3	n/a	50.46	68.9	n/a
zero border	57.79	75.9	54.38	50.52	74.5	49.64
learned operator	57.73	77.1	n/a	50.57	73.6	n/a
convolutional mixer	58.02	85.2	–	50.60	78.5	–
LR-HSI only	39.59	61.9	n/a	30.38	58.4	n/a
HR-MSI only	54.86	59.6	n/a	48.22	53.6	n/a

Which assumptions hold for the trained network. Theorem 2(a)–(b) and Proposition 1 hold for every value of the learned weights: the step sizes are positive by construction (softplus) and the learned gains lie in $(\frac{1}{2}, 2)$ by construction, so the trained data steps keep the ground truth fixed and are nonexpansive. Part (c) needs priors with $\mathcal{P}_k(\mathbf{X}^*) = \mathbf{0}$ and a Lipschitz constant L_k, which training does not enforce; we therefore do not claim convergence of the unrolled network, and report instead that 99.9% of ExactFuse's error lies in the fusion null space (Tab. 1), i.e., its output keeps the observed part.

Cost. Table S13 measures the whole pipeline on a T4. The operator estimate takes 0.08 s (isotropic grid), 0.60 s (wide family) or 5 ms (nonparametric kernel), the eigendecompositions 0.07 s once per image size, and the network 0.24 s. The closed-form step costs about as much as a gradient step (0.21 s), which reproduces the LR-HSI 6.7 dB worse (Tab. 4), and less than a conjugate-gradient step (0.33 and 0.45 s with 5 and 10 iterations). The time grows about linearly with the number of pixels (5.30 s at 1024×1024, against 1.93 s for MIMFormer) and slowly with the number of bands.

Table S6. Where the margin comes from (4 h on one T4). MIMFormer and PSRT with their authors' training recipe and with ours; ExactFuse without its data steps.

	Chikusei		Pavia Centre	
Model and training	PSNR	SAM	PSNR	SAM
MIMFormer, common recipe	56.96	0.801	49.63	1.982
MIMFormer, authors' recipe	56.00	0.874	49.32	2.022
MIMFormer, our recipe	57.52	0.760	50.23	1.954
PSRT, common recipe	53.99	1.057	47.87	2.212
PSRT, authors' recipe	49.51	1.551	45.44	2.597
our prior alone (one U-Net)	57.67	0.747	50.39	1.934
ExactFuse without data steps	57.42	0.766	50.30	1.942
ExactFuse, baselines' recipe	57.67	0.749	50.07	1.921
ExactFuse	58.05	0.724	50.63	1.892

One network for any sensor and scale factor. Every input of the prior lies in the hyperspectral bands, so one ExactFuse network can be trained on random multispectral responses, a new one for every batch, and applied to any sensor (Tab. S9). Tested on seven real sensors, none of them seen in training, it exceeds GSA by 9.9–13.5 dB on Chikusei and 10.5–15.6 dB on Pavia Centre. MIMFormer trained on the same random responses with the same input $\mathcal{R}^+\mathbf{Y}_\mathrm{M}$ stays 3.2–11.3 dB (Chikusei) and 3.6–7.2 dB (Pavia Centre) below it; unlike MIMFormer, ExactFuse uses the response of the sensor at hand in its data steps. Not retraining has

Table S7. All metrics of the equal-budget benchmark (single pass; SSIM with the data range set to each image's maximum; RMSE in DN; –: run not available).

Method	PSNR	SSIM	SAM	ERGAS	CC	SCC	Q2n	RMSE
Chikusei ×4								
Bicubic	33.63	0.7404	4.406	9.011	0.8408	0.7648	0.6914	321.81
GSA	43.20	0.9778	2.730	3.279	0.9905	0.9922	0.9596	107.69
SSR-NET	55.55	0.9958	0.954	1.707	0.9948	0.9987	0.9901	25.30
PSRT	53.99	0.9948	1.057	1.794	0.9942	0.9982	0.9887	30.57
DCT	53.18	0.9945	1.133	2.066	0.9930	0.9978	0.9871	33.58
MIMFormer	56.96	0.9964	0.801	1.488	0.9960	0.9989	0.9918	21.65
DHIF-Net	54.39	0.9953	1.020	1.660	0.9951	0.9983	0.9900	29.37
U2Net	53.35	0.9943	1.076	1.948	0.9937	0.9977	0.9863	32.92
MIMO-SST	57.27	0.9965	0.780	1.419	0.9963	0.9990	0.9922	20.90
OTPNet	57.06	0.9965	0.786	1.448	0.9962	0.9990	0.9922	21.39
SMGU-Net	55.14	0.9957	0.949	1.705	0.9947	0.9986	0.9898	26.60
CLSNet	56.65	0.9963	0.811	1.514	0.9959	0.9989	0.9917	22.35
CaFormer	53.56	0.9949	1.054	2.138	0.9926	0.9980	0.9880	32.07
ExactFuse	58.05	0.9969	0.724	1.334	0.9967	0.9992	0.9930	19.21
Pavia Centre ×4								
Bicubic	25.52	0.6282	6.554	9.852	0.8174	0.7526	0.6806	436.32
GSA	34.78	0.9438	5.229	3.494	0.9958	0.9922	0.9276	157.93
SSR-NET	48.45	0.9954	2.443	0.758	0.9991	0.9991	0.9719	30.46
PSRT	47.87	0.9958	2.212	0.801	0.9990	0.9989	0.9716	32.60
DCT	46.94	0.9953	2.442	0.888	0.9988	0.9985	0.9712	36.26
MIMFormer	49.63	0.9967	1.982	0.674	0.9993	0.9993	0.9738	26.66
DHIF-Net	47.67	0.9956	2.295	0.840	0.9989	0.9988	0.9704	33.42
U2Net	47.29	0.9955	2.327	0.858	0.9989	0.9987	0.9708	34.96
MIMO-SST	49.69	0.9967	1.961	0.670	0.9993	0.9993	0.9741	26.48
OTPNet	50.08	0.9968	1.951	0.653	0.9993	0.9994	0.9737	25.29
SMGU-Net	48.03	0.9958	2.331	0.783	0.9990	0.9990	0.9723	32.05
CLSNet	49.50	0.9967	1.993	0.684	0.9993	0.9993	0.9735	27.09
CaFormer	48.43	0.9961	2.145	0.757	0.9991	0.9991	0.9724	30.56
ExactFuse	50.63	0.9970	1.892	0.619	0.9994	0.9995	0.9753	23.70

a price: on WorldView-2 the sensor-independent network is 1.79 dB below ExactFuse trained for that sensor (1.23 dB on IKONOS for Pavia Centre) and 0.69–0.72 dB below MIMFormer trained for WorldView-2 or Sentinel-2. Trained for one sensor, ExactFuse stays ahead of MIMFormer trained for the same sensor, also with a three-band RGB camera (47.15 against 46.03 dB on Chikusei), where the MSI measures the least. On six sensors with measured responses (6S filter functions, again simulated pairs), the network exceeds MIMFormer trained the same way by 3.1–12.1 dB on Chikusei and 4.0–12.6 dB on Pavia Centre. The data steps also carry the scale factor, so ExactFuse can be trained over several of them, each batch with its own decimation and blur. One network trained at ×4 and ×8 is at most 0.14 and 0.15 dB below the networks trained for each scale (Chikusei, Pavia Centre) and still above the best retrained network of every scale; at ×16, which it never saw, it leads that network by 1.76 and 1.99 dB.

Comparison with reported numbers. Table S16 places ExactFuse trained for 8 h on Chikusei (one run) next to the 17 methods tabulated in [5]; Tab. S17 does the same for the long-trained base model on Pavia Centre. These tables give context, not a ranking: the other rows are copied, our test images follow the published protocol but are not necessarily the same pixels, and our retrained baselines exceed their reported values by up to 3.6 dB (*e.g.* MIMFormer 56.96 dB against 53.36 dB reported). The PSNR values of [5] use a fixed peak: for every row of their Chikusei table, RMSE $\cdot\, 10^{\mathrm{PSNR}/20}$ lies between 16.6k and 17.1k DN, above our peak of 15,133 DN, so our PSNR is conservative by about 0.86 dB on this dataset. RMSE in DN does not depend on the peak and is the most direct comparison.

S8. Open problems and future directions

Most fusion methods, ours included in its benchmark setting, assume co-registered inputs, white noise of one level, a known spatially invariant blur and simultaneous acquisitions. We list the gaps that remain and how the exact-operator view gives each a concrete entry point.

Registered and unregistered pixels. Real pairs are misregistered locally (terrain and building parallax, moving objects, different view angles), so one image contains registered and unregistered pixels. A global sub-pixel offset fits into the separable operator and is estimated from the pair with the criterion of Theorem 3; a local displacement does not. The confidence-weighted data step treats unregistered pixels as unreliable measurements and its map $1 - w$ marks them, which turns a nuisance into an output; a spatially varying operator (a displacement field estimated jointly with the image) is the principled next step.

Realistic noise. Hyperspectral noise depends on the band (low SNR at the band edges and in absorption bands) and on the signal (photon noise), and pushbroom sensors add stripes. Per-band noise levels enter the MAP derivation as per-band weights of the data term; with the imputation form of the confidence step they keep the closed form. Training and evaluation with band- and signal-dependent noise, and with stripes, should replace the white-noise protocol.

Spatially varying and unknown blur. Real point-spread functions vary over the field of view and are rarely Gaussian. Non-separable kernels keep the gradient and conjugate-gradient steps but lose the closed form; estimating a non-parametric kernel from $\mathcal{D}\mathbf{Y}_{\mathrm{M}} = \mathcal{R}\mathbf{Y}_{\mathrm{H}}$ (Theorem 3) and learning operator families from calibration data are open.

Changes between acquisition dates. Hyperspectral and multispectral images of the same scene are often taken days apart (e.g., EnMAP or PRISMA with Sentinel-2). Classical robust fusion models the change as a sparse outlier [8]; the confidence step is a learned counterpart whose map is a change detector, and joint fusion and change detection on real multi-date pairs is a natural application.

Real references and trustworthy outputs. Wald's protocol simulates both inputs, so it cannot measure robustness to the real operator. Real paired benchmarks (e.g., airborne spectrometer and camera flights, or lens-based capture), no-reference metrics that respect the null space, and calibrated uncertainty for the part of the image that only a prior supplies are needed before fused products are used for quantitative analysis.

Other inverse problems with Kronecker structure. The same operators describe hyperspectral pansharpening ($M = 1$), spectral super-resolution from RGB (no $\mathcal{D}$), multispectral pansharpening and multi-temporal or multi-sensor fusion; the closed-form data step, the null-space split and the band-space interface carry over to each.

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Table S8. All metrics of the equal-budget benchmark on the other datasets and scale factors (single pass; SSIM with the data range set to each image's maximum; RMSE in DN, on the 8-bit scale for CAVE and Harvard; –: run not available).

Method	PSNR	SSIM	SAM	ERGAS	CC	SCC	Q2n	RMSE
CAVE ×4								
Bicubic	30.05	0.8780	4.795	8.229	0.9679	0.7220	0.7518	8.67
GSA	42.30	0.9798	4.651	2.080	0.9981	0.9849	0.8949	2.13
SSR-NET	46.51	0.9942	2.922	1.254	0.9989	0.9938	0.9577	1.29
PSRT	48.51	0.9950	2.394	1.016	0.9993	0.9950	0.9675	1.03
DCT	46.36	0.9934	2.794	1.290	0.9990	0.9928	0.9579	1.31
MIMFormer	50.93	0.9966	1.955	0.786	0.9996	0.9966	0.9718	0.78
DHIF-Net	49.82	0.9962	2.047	0.896	0.9995	0.9959	0.9708	0.90
U2Net	48.04	0.9941	2.443	1.110	0.9992	0.9943	0.9614	1.10
MIMO-SST	50.39	0.9963	2.026	0.829	0.9995	0.9958	0.9721	0.83
SMGU-Net	48.54	0.9952	2.311	1.025	0.9993	0.9947	0.9682	1.03
CaFormer	47.91	0.9950	2.459	1.110	0.9993	0.9951	0.9632	1.11
ExactFuse	52.46	0.9972	1.659	0.664	0.9997	0.9975	0.9744	0.65
Harvard ×4								
Bicubic	35.53	0.8588	2.690	4.543	0.9415	0.7894	0.6642	4.69
GSA	46.30	0.9766	2.379	1.669	0.9917	0.9879	0.8695	1.34
SSR-NET	49.18	0.9813	2.148	1.387	0.9920	0.9913	0.8911	0.95
PSRT	49.29	0.9813	2.127	1.344	0.9925	0.9914	0.8921	0.93
DCT	48.92	0.9808	2.188	1.441	0.9917	0.9911	0.8881	0.98
MIMFormer	49.72	0.9822	2.061	1.287	0.9931	0.9923	0.8960	0.88
DHIF-Net	49.58	0.9820	2.084	1.301	0.9930	0.9920	0.8947	0.90
U2Net	48.70	0.9793	2.224	1.666	0.9880	0.9909	0.8759	1.00
MIMO-SST	49.83	0.9826	2.039	1.276	0.9932	0.9926	0.8987	0.87
SMGU-Net	49.34	0.9815	2.122	1.353	0.9923	0.9916	0.8925	0.93
CaFormer	49.09	0.9811	2.177	1.403	0.9919	0.9917	0.8885	0.95
ExactFuse	50.03	0.9830	2.009	1.232	0.9937	0.9933	0.9025	0.84
CAVE ×8								
Bicubic	26.05	0.7945	6.781	6.414	0.9197	0.5647	0.5802	13.70
GSA	38.01	0.9531	7.273	1.725	0.9967	0.9815	0.8199	3.58
SSR-NET	44.25	0.9918	3.657	0.793	0.9982	0.9928	0.9513	1.68
PSRT	45.21	0.9916	3.229	0.708	0.9986	0.9929	0.9570	1.52
DCT	43.17	0.9883	3.929	0.891	0.9981	0.9900	0.9414	1.89
MIMFormer	48.10	0.9950	2.413	0.539	0.9992	0.9952	0.9679	1.11
DHIF-Net	–	–	–	–	–	–	–	–
ExactFuse	50.39	0.9962	1.934	0.429	0.9995	0.9962	0.9715	0.86
Harvard ×8								
Bicubic	–	–	–	–	–	–	–	–
GSA	–	–	–	–	–	–	–	–
SSR-NET	48.61	0.9805	2.262	0.719	0.9914	0.9904	0.8879	1.04
PSRT	–	–	–	–	–	–	–	–
DCT	–	–	–	–	–	–	–	–
MIMFormer	–	–	–	–	–	–	–	–
DHIF-Net	–	–	–	–	–	–	–	–
ExactFuse	–	–	–	–	–	–	–	–
Chikusei ×8								
Bicubic	31.20	0.6503	6.127	5.980	0.7034	0.6287	0.4389	425.27
GSA	38.69	0.9524	3.972	2.571	0.9847	0.9839	0.9069	180.29
SSR-NET	54.57	0.9953	1.030	0.926	0.9938	0.9985	0.9885	28.35
PSRT	53.05	0.9941	1.172	1.004	0.9927	0.9979	0.9863	33.95
DCT	51.77	0.9934	1.318	1.198	0.9905	0.9973	0.9839	39.69
MIMFormer	55.57	0.9957	0.919	0.862	0.9946	0.9987	0.9897	25.33
DHIF-Net	–	–	–	–	–	–	–	–
MIMO-SST	55.88	0.9958	0.895	0.828	0.9949	0.9987	0.9901	24.44
OTPNet	56.11	0.9960	0.871	0.816	0.9951	0.9988	0.9904	23.83
ExactFuse	56.95	0.9963	0.817	0.755	0.9956	0.9990	0.9912	21.78
Pavia Centre ×8								
Bicubic	23.07	0.4775	8.499	6.541	0.6499	0.6020	0.4060	575.39
GSA	30.24	0.8731	7.163	2.960	0.9927	0.9841	0.8416	269.22
SSR-NET	47.71	0.9951	2.648	0.403	0.9990	0.9990	0.9713	33.17
PSRT	47.13	0.9953	2.394	0.430	0.9988	0.9987	0.9706	35.49
DCT	45.51	0.9941	2.739	0.521	0.9983	0.9981	0.9697	42.82
MIMFormer	49.09	0.9964	2.064	0.353	0.9992	0.9992	0.9729	28.38
DHIF-Net	–	–	–	–	–	–	–	–
MIMO-SST	48.90	0.9964	2.090	0.360	0.9992	0.9992	0.9731	29.03
OTPNet	49.51	0.9966	2.016	0.342	0.9993	0.9993	0.9730	26.97
ExactFuse	50.05	0.9967	1.977	0.327	0.9993	0.9994	0.9739	25.33
Chikusei ×16								
Bicubic	29.77	0.6186	7.611	3.523	0.5555	0.5163	0.2136	500.32
GSA	35.20	0.9072	5.303	1.860	0.9781	0.9637	0.8093	268.91
SSR-NET	53.98	0.9949	1.086	0.501	0.9928	0.9984	0.9874	30.32
PSRT	–	–	–	–	–	–	–	–
DCT	–	–	–	–	–	–	–	–
MIMFormer	54.21	0.9951	1.029	0.490	0.9931	0.9984	0.9878	29.54
DHIF-Net	–	–	–	–	–	–	–	–
ExactFuse	56.11	0.9960	0.882	0.408	0.9949	0.9988	0.9901	23.87
Pavia Centre ×16								
Bicubic	–	–	–	–	–	–	–	–
GSA	–	–	–	–	–	–	–	–
SSR-NET	47.38	0.9945	2.662	0.207	0.9990	0.9989	0.9709	34.39
PSRT	45.47	0.9935	2.755	0.264	0.9983	0.9981	0.9679	42.89
DCT	43.43	0.9921	3.177	0.334	0.9973	0.9975	0.9663	54.37
MIMFormer	–	–	–	–	–	–	–	–
DHIF-Net	–	–	–	–	–	–	–	–
ExactFuse	–	–	–	–	–	–	–	–

Table S9. One network for any multispectral sensor and for several scale factors (test PSNR, dB; Chikusei and Pavia Centre). Top: networks with the sensor-independent input $\mathcal{R}^{+}\mathbf{Y}_{\mathrm{M}}$ trained on random responses (3–10 bands of random position and width, drawn anew for every batch) and tested on the responses of seven real sensors, none of which was seen in training (the pairs are simulated from the reference with these responses, not acquired by the sensors; M: bands inside the hyperspectral range), against GSA and against networks trained for that one sensor (Retr.). Bottom: ExactFuse trained over $\times4$ and $\times8$ and tested at $\times4$, $\times8$ and $\times16$, against ExactFuse trained at each scale and the best retrained network of that scale. –: not run. Best of the two sensor-independent networks in bold.

	Chikusei $\times4$						Pavia Centre $\times4$					
			One network		Retr. for sensor				One network		Retr. for sensor	
Sensor	M	GSA	MIMFormer	ExactFuse	MIMFormer	ExactFuse	M	GSA	MIMFormer	ExactFuse	MIMFormer	ExactFuse
WorldView-2	8	43.20	46.17	**56.26**	56.96	58.05	6	34.69	46.37	**50.28**	–	–
IKONOS	4	42.64	50.33	**53.57**	–	–	4	34.78	45.55	**49.39**	49.63	50.63
QuickBird	4	42.70	50.74	**53.93**	–	–	4	34.66	45.80	**49.35**	–	–
PlanetScope	4	40.64	44.33	**53.98**	–	–	3	29.82	33.76	**40.95**	–	–
Landsat 8	5	40.67	42.85	**54.12**	–	–	3	29.82	34.00	**40.77**	–	–
Sentinel-2	10	43.43	49.47	**56.06**	56.79	–	7	35.24	43.45	**50.52**	–	–
RGB camera	3	34.49	36.92	**44.41**	46.03	47.15	3	29.77	35.15	**40.30**	–	–

Scale	GSA	ExactFuse, $\times4$+$\times8$	ExactFuse	best retr.	GSA	ExactFuse, $\times4$+$\times8$	ExactFuse	best retr.
$\times4$	43.20	57.96	58.05	57.27	34.78	50.47	50.63	50.08
$\times8$	38.69	56.89	56.95	56.11	30.24	49.96	50.05	49.51
$\times16$	35.20	55.97	56.11	54.21	27.07	49.38	–	47.38

Table S10. Conditioning and stability of the operator estimate (Theorem 3(c)) on the test images. $\tilde{\kappa}$: condition number of the Jacobian $\mathbf{G}$ of the residual with respect to the five parameters (two blur widths, two grid offsets, response shift) after scaling its columns to unit norm. pred.: first-order standard deviation $\mathbf{G}^{+}\boldsymbol{\Sigma}_{\mathbf{n}}\mathbf{G}^{+\top}$; meas.: RMSE of the estimate (grid search refined by Levenberg–Marquardt) over the test images and three noise draws. Blur width and offset in pixels (mean of the two axes), response shift in nm.

			40 dB SNR						30 dB SNR						20 dB SNR					
			blur		offset		shift		blur		offset		shift		blur		offset		shift	
Dataset	Case	$\tilde{\kappa}$	pred.	meas.	pred.	meas.	pred.	meas.	pred.	meas.	pred.	meas.	pred.	meas.	pred.	meas.	pred.	meas.	pred.	meas.
Chikusei	nominal	1.28	0.0005	0.0008	0.0004	0.0005	0.003	0.003	0.0017	0.0042	0.0013	0.0013	0.010	0.013	0.0054	0.0363	0.0042	0.0038	0.030	0.033
	anisotropic	1.23	0.0007	0.0010	0.0005	0.0006	0.003	0.003	0.0024	0.0050	0.0016	0.0014	0.010	0.013	0.0075	0.0417	0.0049	0.0048	0.030	0.033
	offsets and shift	1.29	0.0006	0.0009	0.0004	0.0005	0.003	0.003	0.0017	0.0041	0.0013	0.0014	0.010	0.012	0.0055	0.0370	0.0042	0.0036	0.032	0.042
Pavia Centre	nominal	1.47	0.0004	0.0004	0.0003	0.0003	0.012	0.022	0.0014	0.0016	0.0010	0.0013	0.038	0.075	0.0043	0.0126	0.0031	0.0040	0.119	0.697
	anisotropic	1.38	0.0006	0.0005	0.0004	0.0003	0.012	0.022	0.0018	0.0020	0.0011	0.0014	0.038	0.078	0.0057	0.0139	0.0036	0.0050	0.119	0.725
	offsets and shift	1.47	0.0004	0.0004	0.0003	0.0003	0.013	0.020	0.0014	0.0016	0.0010	0.0013	0.042	0.066	0.0043	0.0133	0.0030	0.0035	0.132	0.560

Table S11. Per-image gain of ExactFuse over the strongest retrained network of each setting (by mean PSNR): mean PSNR gain (dB) with its 95% paired bootstrap interval over the test images (10,000 resamples), and the number of test images on which ExactFuse has the higher PSNR and the lower SAM and ERGAS.

Setting	Strongest	n	Gain	95% CI	PSNR	SAM	ERGAS
Chik. ×4	MIMO-SST	8	0.79	[0.52, 1.08]	8/8	8/8	8/8
Pavia ×4	OTPNet	4	0.55	[0.41, 0.69]	4/4	4/4	4/4
CAVE ×4	MIMFormer	12	1.53	[1.30, 1.75]	12/12	12/12	12/12
Harv. ×4	MIMO-SST	20	0.21	[0.11, 0.33]	19/20	20/20	20/20
CAVE ×8	MIMFormer	12	2.29	[1.82, 2.81]	12/12	12/12	12/12
Chik. ×8	OTPNet	8	0.84	[0.51, 1.22]	8/8	8/8	8/8
Pavia ×8	OTPNet	4	0.53	[0.42, 0.75]	4/4	4/4	4/4
Chik. ×16	MIMFormer	8	1.90	[1.18, 2.74]	8/8	8/8	8/8

Table S12. Every test image: PSNR (dB), SAM (degrees) and ERGAS of ExactFuse and of the strongest retrained network; the better value of each pair in bold.

Image	PSNR		SAM		ERGAS	
Chikusei ×4, ExactFuse / MIMO-SST						
1	**58.86**	57.69	**0.664**	0.737	**1.562**	1.682
2	**60.00**	58.62	**0.671**	0.747	**0.994**	1.088
3	**60.67**	59.35	**0.615**	0.686	**1.104**	1.235
4	**57.01**	56.46	**0.874**	0.935	**1.340**	1.423
5	**57.24**	56.82	**0.731**	0.768	**1.303**	1.348
6	**57.43**	56.99	**0.675**	0.711	**1.361**	1.421
7	**56.35**	55.78	**0.823**	0.876	**1.404**	1.501
8	**56.83**	56.41	**0.741**	0.780	**1.602**	1.655
Pavia Centre ×4, ExactFuse / OTPNet						
1	**50.37**	49.69	**1.560**	1.608	**0.559**	0.594
2	**52.28**	51.92	**2.714**	2.795	**0.815**	0.844
3	**49.43**	48.72	**1.690**	1.741	**0.562**	0.602
4	**50.43**	49.98	**1.602**	1.661	**0.541**	0.571

Table S13. End-to-end cost on one Tesla T4 (batch 1, Chikusei bands and a 256×256 HR-MSI unless stated; seconds per image and peak GPU memory). The data-step rows use the same network with another data step.

Step	Time (s)	Memory (GB)
Operator estimate, isotropic grid	0.081	–
Operator estimate, wide family	0.600	–
Operator estimate, nonparametric kernel	0.0048	–
Eigendecompositions (once per image size)	0.069	–
Network, closed-form step with learned gains	0.241	0.54
Network, closed form	0.242	0.54
Network, gradient step	0.206	0.60
Network, conjugate gradient, 5 iterations	0.333	0.54
Network, conjugate gradient, 10 iterations	0.447	0.54
128×128: ExactFuse / MIMFormer / eigendecompositions	0.090 / 0.053 / 0.007	0.17 / 0.17
256×256: ExactFuse / MIMFormer / eigendecompositions	0.242 / 0.115 / 0.022	0.54 / 0.46
512×512: ExactFuse / MIMFormer / eigendecompositions	1.192 / 0.445 / 0.071	2.02 / 1.58
1024×1024: ExactFuse / MIMFormer / eigendecompositions	5.301 / 1.931 / 0.374	7.92 / 6.11
31 bands (256×256, $M = 8$): ExactFuse	0.187	0.29
64 bands (256×256, $M = 8$): ExactFuse	0.197	0.38
128 bands (256×256, $M = 8$): ExactFuse	0.244	0.56
224 bands (256×256, $M = 8$): ExactFuse	0.384	0.92

Table S14. The sensor-independent networks of Tab. S9 (trained on random responses) on six sensors with measured spectral responses (6S filter functions), none seen in training; the pairs are simulated from the reference with these responses, not acquired by the sensors; M: bands inside the hyperspectral range. Test PSNR (dB).

	Chikusei				Pavia Centre			
Sensor	M	GSA	MIMF.	ExactFuse	M	GSA	MIMF.	ExactFuse
Landsat 8 OLI	5	35.55	42.01	**54.07**	3	28.31	33.47	**40.73**
Sentinel-2A MSI	10	38.82	52.56	**55.89**	7	32.78	39.28	**50.19**
Sentinel-2B MSI	10	38.79	52.24	**55.87**	7	32.82	39.68	**50.23**
Sentinel-3 OLCI	21	37.07	49.67	**56.39**	13	29.50	35.61	**47.91**
PROBA-V	3	38.21	50.57	**53.69**	2	33.22	40.75	**44.74**
MODIS (Aqua)	11	36.15	44.75	**54.34**	9	29.84	36.31	**48.88**

Table S15. HR-MSI misregistered against the LR-HSI and the reference (test PSNR, dB). Registered: the HR-MSI is shifted back by the offset estimated from the pair before fusion (any method). Bilinear: the shift of the simulation also blurs the HR-MSI; sharp: band-limited shift, as a sensor sampling a shifted scene records it, registered with the same interpolation. ExactFuse is trained on noise-free data, the wider family adds noise, anisotropic blurs and grid offsets; MIMFormer is trained on noise-free data. Best registered result in bold.

	Chikusei ×4					Pavia Centre ×4				
Misregistration	ExactFuse, unreg.	ExactFuse	ExactFuse, wider family	MIMFormer	GSA	ExactFuse, unreg.	ExactFuse	ExactFuse, wider family	MIMFormer	GSA
0.5–1 px (bilinear)	37.88	**49.74**	49.59	49.38	42.42	29.72	**40.66**	40.58	40.38	33.78
0.5 px + 35 dB	39.91	44.39	**45.87**	45.61	41.67	32.14	36.67	**37.14**	37.05	33.16
1 px + 35 dB	34.83	46.89	**50.46**	49.18	42.43	27.01	41.12	**42.76**	42.10	34.12
0.5 px, sharp	39.87	**54.56**	54.41	54.05	43.06	31.43	**46.36**	46.34	45.97	34.59
0.5 px, sharp + 35 dB	39.20	47.26	**51.57**	49.98	42.59	31.25	42.24	**44.79**	43.83	34.40
0.5 px + aniso. blur + 35 dB	39.53	43.48	**45.72**	45.49	41.59	31.55	35.12	36.90	**36.92**	33.11

Table S16. Chikusei ×4: ExactFuse trained for 8 h next to the methods reported in [5] (all other rows copied from there). SSIM as in the PSRT code (data range 1).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [1]	47.5769	0.9933	1.7216	2.1667	0.9658	0.9894	0.9773	70.54
CNMF [25]	44.5291	0.9891	1.7259	2.8676	0.9662	0.9828	0.9645	101.64
FUSE [20]	41.1550	0.9787	2.5814	3.4710	0.9418	0.9736	0.9486	148.10
U2Net [18]	50.5455	0.9956	1.2389	1.8740	0.9928	0.9920	0.9798	49.97
PSRT [6]	50.6722	0.9952	1.2733	2.0107	0.9862	0.9906	0.9778	49.00
DCT [17]	52.1998	0.9977	0.9239	1.7448	0.9832	0.9935	0.9873	41.45
DDIF [3]	53.4421	0.9972	0.9387	1.8087	0.9963	0.9929	0.9840	35.58
MIMFormer [13]	53.3578	0.9973	0.9094	1.7611	0.9948	0.9931	0.9850	35.90
MIMO-SST [7]	53.6073	0.9972	0.9315	1.8119	0.9944	0.9926	0.9834	34.93
OTPNet [26]	52.8243	0.9969	1.0201	1.9332	0.9940	0.9916	0.9841	38.49
SMGU [21]	54.5244	0.9977	0.8719	1.6405	0.9952	0.9939	0.9875	31.29
AMSF [15]	49.3827	0.9944	1.4852	2.3943	0.9878	0.9879	0.9790	56.78
SSDT [9]	52.4788	0.9975	0.8768	1.7992	0.9939	0.9928	0.9847	40.00
ASMNet [14]	53.4528	0.9972	0.9472	1.9153	0.9874	0.9917	0.9848	35.89
DPFormer [16]	53.1632	0.9970	0.9483	1.8795	0.9949	0.9923	0.9824	37.18
CLSNet [22]	55.2007	0.9979	0.7836	1.6318	0.9943	0.9941	0.9869	29.06
Two-stage diffusion [5]	56.1871	0.9982	0.7203	1.5158	**0.9969**	0.9948	0.9891	25.73
ExactFuse, 8 h (single pass)	58.1840	**0.9989**	0.7146	1.3284	0.9931	**0.9967**	**0.9992**	18.96
ExactFuse, 8 h (self-ensemble)	**58.2129**	**0.9989**	**0.7129**	**1.3260**	0.9931	**0.9967**	**0.9992**	**18.91**

Table S17. Pavia Centre ×4 comparison. All rows except the base model are reported in [5] on the benchmark data released with PSRT [6]; the base model is evaluated on our simulation of the same protocol from the original scene (Section S3). Best value per column in bold. SSIM of the base model uses the PSRT definition (data range 1); with data range equal to each image's maximum it is 0.9970 (single pass) and 0.9970 (self-ensemble).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [1]	35.3495	0.9736	5.0812	2.7260	0.9818	0.9832	0.9768	137.78
CNMF [25]	32.1693	0.9469	3.8494	3.8446	0.9489	0.9683	0.9294	198.26
FUSE [20]	34.7418	0.9612	5.5628	3.2042	0.9103	0.9779	0.9469	146.90
U2Net [18]	44.3439	0.9963	1.6542	1.0924	0.9964	0.9973	0.9963	49.10
PSRT [6]	45.8875	0.9970	1.5230	0.9750	0.9974	0.9978	0.9966	41.85
DCT [17]	46.6745	0.9971	1.5169	0.8753	0.9973	0.9983	0.9972	37.66
DDIF [3]	46.2530	0.9970	1.5253	0.9125	0.9960	0.9982	0.9970	39.56
MIMFormer [13]	44.8497	0.9959	1.8225	1.0683	0.9957	0.9975	0.9954	46.19
MIMO-SST [7]	46.6334	0.9969	1.5587	0.8607	0.9975	0.9984	0.9972	37.68
OTPNet [26]	45.8081	0.9967	1.6573	0.9465	**0.9987**	0.9980	0.9968	41.51
SMGU [21]	47.1239	0.9972	1.4690	0.8547	0.9981	0.9984	0.9974	37.09
AMSF [15]	43.4465	0.9950	2.0533	1.2728	0.9929	0.9965	0.9940	54.25
SSDT [9]	46.3173	0.9965	1.6439	0.8918	0.9968	0.9982	0.9969	39.01
ASMNet [14]	46.2072	0.9966	1.6517	0.9019	0.9940	0.9982	0.9969	39.50
DPFormer [16]	42.4280	0.9948	1.9367	1.3986	0.9945	0.9957	0.9917	60.96
CLSNet [22]	46.5277	0.9969	1.5708	0.8412	0.9983	0.9983	0.9975	36.21
Two-stage diffusion [5]	47.1597	0.9972	**1.4542**	0.8262	0.9980	0.9985	0.9976	35.71
Base model (single pass)	50.6299	**0.9974**	1.8955	0.6206	0.9751	**0.9994**	**0.9995**	23.68
Base model (self-ensemble)	**50.6915**	**0.9974**	1.8833	**0.6170**	0.9753	**0.9994**	**0.9995**	**23.52**

Table S18. Real data (NEON, SERC 2019: imaging spectrometer, 120 bands 400–1000 nm, with the measured camera RGB). Reduced resolution: the LR-HSI is simulated from the 1 m HSI (×4), the RGB is measured and the 1 m HSI is the reference. Full resolution: the measured 1 m HSI is fused with the measured 0.25 m RGB, without a reference. All networks retrained under the same 4 h budget; best retrained network in bold.

	Reduced resolution (reference)				Full resolution (no reference)		
Method	PSNR↑	SAM↓	ERGAS↓	SSIM↑	D_λ↓	D_s↓	HQNR↑
Bicubic	31.78	3.422	13.679	0.7779	0.1966	0.0659	0.7494
GSA	31.25	6.087	14.115	0.7414	0.4259	0.1880	0.4774
SSR-NET [28]	36.37	2.707	7.794	0.9042	–	–	–
PSRT [6]	36.74	2.434	7.483	0.9111	0.1191	0.0514	0.8366
DCT [17]	36.43	2.816	7.841	0.9037	0.1533	0.0666	0.7938
MIMFormer [13]	38.45	2.075	5.893	0.9342	0.0635	0.0465	0.8923
DHIF-Net [11]	36.97	2.357	7.119	0.9129	0.0854	**0.0388**	0.8783
ExactFuse	**38.70**	**1.852**	**5.577**	**0.9394**	**0.0209**	0.0599	**0.9203**
blur + noise randomized	38.62	1.890	5.624	0.9385	–	–	–
base model, zero border	38.68	1.868	5.601	0.9394	0.0278	0.0629	0.9109
base model, learned operator	38.68	1.867	5.562	0.9393	0.0266	0.0651	0.9099
base model, no data step	38.63	1.897	5.640	0.9389	0.0555	0.0569	0.8905

Table S19. Second NEON site: networks trained on SERC (Maryland) and tested, without retraining, on SJER (California, 2019, 20 test crops of five tiles; the camera response is fitted on three other SJER tiles). PSNR in dB at reduced resolution, as in Tab. S18.

	SERC	SJER (unseen site)		
Method	PSNR↑	PSNR↑	SAM↓	ERGAS↓
Bicubic	–	30.55	–	–
SSR-NET [28]	36.37	34.60	2.39	4.79
PSRT [6]	36.74	35.53	1.88	4.34
DCT [17]	36.43	35.35	2.00	4.44
DHIF-Net [11]	36.97	35.71	1.80	3.98
ExactFuse	38.70	36.91	1.46	3.45
trained over the operator family	38.62	36.66	1.52	3.65

Table S20. Ablation: each variant changes one part of the full model and is trained from scratch with the same 4 h budget on one T4 (test set, single pass). Δ: PSNR change against the full model. Cons.: PSNR$(\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_{\mathrm{H}})$ in dB. Swap: worst test PSNR over the blurs $\sigma \in \{0.75, 1.25, 1.5\}\sigma_0$ with the true blur given at test time; n/a: no operator to replace. $\pm$: mean and standard deviation over three seeds.

			Pavia Centre $\times 4$				Chikusei $\times 4$			
#	Variant	Params (M)	PSNR	Δ	Cons.	Swap	PSNR	Δ	Cons.	Swap
(1)	base model (full)	9.99	50.62$\pm$0.02	–	77.6	50.50	57.95$\pm$0.004	–	83.9	57.70
(2)	no data step, no residual input	9.99	50.46	-0.17	68.9	n/a	57.65	-0.30	73.3	n/a
(3)	no residual input	9.99	50.62	-0.01	75.7	–	58.01	+0.06	82.8	–
(4)	exact kernel, zero border	9.99	50.54$\pm$0.03	-0.08	74.5	49.64	57.73$\pm$0.05	-0.22	75.9	54.38
(5)	learned blur kernel and SRF	9.99	50.57	-0.06	73.6	n/a	57.73	-0.22	77.1	n/a
(6)	exact proximal step, conjugate gradients	9.99	50.55	-0.07	79.4	50.44	57.74	-0.21	85.2	57.54
(7)	exact proximal step, closed form	9.99	50.65	+0.02	80.9	50.58	57.93	-0.02	87.3	57.76
(8)	learned spectral data step	9.99	50.63	+0.01	82.9	50.50	58.00	+0.05	89.4	57.83
(9)	with the HR-MSI residual input	10.11	50.64	+0.01	79.0	50.58	58.04	+0.09	86.5	57.85
(10)	no cross-stage memory	9.98	50.67	+0.05	77.2	–	57.96	+0.01	84.1	–
(11)	$K = 1$ stage	3.33	50.60	-0.02	71.3	–	57.75	-0.20	77.4	–
(12)	$K = 2$ stages	6.66	50.64	+0.02	75.0	–	57.92	-0.02	81.0	–
(13)	$K = 4$ stages	13.32	50.68	+0.06	80.2	–	57.96	+0.01	85.0	–
(14)	convolutional mixer	9.94	50.60	-0.02	78.5	–	58.02	+0.07	85.2	–
(15)	8×8 window attention	9.99	50.63	+0.01	78.6	–	57.98	+0.03	84.4	–
(16)	L1 + deep supervision only	9.99	50.66	+0.03	77.5	–	57.97	+0.02	82.2	–
(17)	no deep supervision	9.99	50.50	-0.12	77.7	–	57.99	+0.04	84.1	–

Table S21. Operator ablation across datasets and scale factors (one run each, seed 0, 4 h on one T4; test PSNR in dB). Training blur: PSNR at the blur used for training. Changed blur: worst PSNR over the three test blurs $\sigma \in \{0.75, 1.25, 1.5\}\,\sigma_0$, with the true blur supplied (swap) or the training operator kept (fixed). –: run not available.

	Training blur				Changed blur (worst case)				
Setting	Exact	Zero border	Learned	No data step	Exact, swap	Exact, fixed	Zero border, swap	Learned	No data step
Chikusei $\times 4$	57.95	57.79	57.73	57.65	57.70	51.41	54.38	54.28	–
Pavia Centre $\times 4$	50.63	50.52	50.57	50.46	50.50	43.13	49.64	49.33	–
CAVE $\times 4$	52.57	52.19	52.22	51.69	52.29	48.66	51.82	51.08	51.38
Harvard $\times 4$	50.01	49.97	49.98	49.84	49.95	48.41	49.29	49.69	49.74
Chikusei $\times 8$	56.82	56.53	56.39	–	56.69	50.00	55.27	55.31	–
Pavia Centre $\times 8$	49.94	49.65	49.56	–	49.88	42.69	49.38	48.43	–

Table S22. Operator-randomized training, every test condition (PSNR in dB; columns as in the operator-family table of the main paper). Est. error: mean absolute error of the estimated blur width and SRF shift (nm) over the test images.

		MIMFormer, nominal training	MIMFormer, randomized	ExactFuse, nominal training		base model, randomized data only		base model, randomized, zero border	base model, randomized + operator			+ observation-model conditioning			ExactFuse, randomized + conditioning		
Condition	Est. error	fixed	fixed	fixed	est.	fixed	est.	est.	fixed	est.	true	fixed	est.	true	fixed	est.	true
Chikusei ×4																	
nominal	0.00/0.0	56.96	56.50	58.05	58.05	57.43	57.43	57.46	57.59	57.59	57.59	57.61	57.61	57.61	57.77	57.77	57.77
sigma x0.5	0.00/0.0	49.85	56.16	40.84	57.30	57.11	56.66	57.19	48.71	57.28	57.28	48.94	57.31	57.31	49.90	57.47	57.47
sigma x0.75	0.00/0.0	55.00	56.50	48.81	57.94	57.39	57.30	57.41	54.57	57.56	57.56	54.54	57.57	57.57	55.16	57.72	57.72
sigma x1.25	0.00/0.0	56.24	56.46	52.87	58.01	57.41	57.34	57.44	56.34	57.57	57.57	56.26	57.59	57.59	56.69	57.77	57.77
sigma x1.5	0.00/0.0	55.39	56.41	49.93	57.93	57.39	57.24	57.41	55.08	57.54	57.54	55.00	57.56	57.56	55.54	57.75	57.75
sigma x2	0.00/0.0	54.41	56.35	47.55	57.79	57.34	57.10	57.35	53.76	57.49	57.49	53.66	57.52	57.52	54.31	57.71	57.71
shift -12 nm	0.00/0.0	51.16	54.56	49.75	53.26	57.27	53.44	57.35	48.09	57.44	57.44	45.91	57.50	57.50	46.45	57.73	57.73
shift -8 nm	0.00/0.0	53.47	55.98	52.47	55.15	57.39	55.24	57.44	51.07	57.55	57.55	49.13	57.59	57.59	49.62	57.78	57.78
shift -4 nm	0.00/0.0	55.82	56.41	55.67	57.04	57.43	56.77	57.47	54.88	57.59	57.59	53.65	57.61	57.61	54.02	57.79	57.79
shift +4 nm	0.00/0.0	55.80	56.36	55.96	56.99	57.40	56.72	57.43	54.89	57.56	57.56	53.70	57.58	57.58	54.15	57.75	57.75
shift +8 nm	0.00/0.0	53.69	55.87	52.61	55.02	57.34	55.17	57.37	51.26	57.51	57.51	49.42	57.52	57.52	49.96	57.71	57.71
shift +12 nm	0.00/0.0	51.74	54.82	49.87	53.29	57.25	53.48	57.27	48.52	57.41	57.41	46.47	57.42	57.42	47.02	57.64	57.64
snr 40 dB	0.00/0.0	54.08	54.54	51.89	51.89	55.03	55.03	55.12	55.31	55.31	55.31	55.36	55.36	55.36	55.29	55.29	55.29
snr 35 dB	0.00/0.0	50.93	52.07	47.76	47.76	52.70	52.70	52.87	53.13	53.13	53.13	53.25	53.25	53.25	52.97	52.97	52.97
snr 30 dB	0.00/0.0	46.00	48.40	43.17	43.17	49.33	49.33	49.58	49.93	49.93	49.93	50.14	50.14	50.14	49.66	49.66	49.66
sigma x1.5, shift +8 nm, snr 35 dB	0.00/0.0	49.66	51.79	45.23	46.54	52.63	51.65	52.81	49.29	53.06	53.06	47.63	53.16	53.16	48.30	52.91	52.91
sigma x0.75, shift -8 nm, snr 40 dB	0.00/0.0	51.49	54.10	46.44	51.71	55.00	53.57	55.05	49.74	55.26	55.26	47.86	55.34	55.34	48.44	55.28	55.28
sigma x2, shift +12 nm, snr 30 dB	0.01/0.0	45.78	48.03	41.25	41.89	49.25	48.12	49.49	45.99	49.80	49.80	44.09	49.98	49.98	44.94	49.57	49.57
aniso blur 1.5/0.75 sigma0, 30 deg	–/0.0	55.65	56.49	49.95	50.14	57.26	57.27	56.84	55.42	55.60	57.46	55.28	55.41	57.47	55.82	55.97	57.68
aniso blur 1.25/0.5 sigma0, 0 deg	–/0.0	52.88	56.33	45.39	47.78	56.98	56.74	55.33	51.96	52.55	57.32	52.01	52.69	57.11	52.82	53.35	57.47
disk blur r = 1.25 sigma0	–/0.0	54.16	56.48	46.82	54.46	57.31	57.18	57.01	53.40	56.22	57.44	53.43	56.24	57.34	54.09	56.71	57.64
MSI misregistered 0.5 px	0.03/0.0	40.03	39.41	40.70	40.70	39.43	39.43	39.81	40.13	40.13	–	40.17	40.17	–	40.13	40.13	–
MSI misregistered 1 px	0.72/0.1	34.35	33.71	35.06	34.93	33.70	33.70	34.03	34.45	34.36	–	34.45	34.36	–	34.39	34.28	–
aniso blur, MSI 0.5 px, snr 35 dB	–/0.0	39.63	39.25	39.53	39.51	39.32	39.32	39.63	39.88	39.88	–	39.83	39.83	–	39.76	39.75	–
HSI grid offset 0.5/0.5 px	–/0.0	53.25	56.52	47.77	47.75	57.17	57.18	–	52.96	53.00	57.57	52.90	52.93	57.58	53.89	53.93	57.79
HSI grid offset 1/-1 px	–/0.0	49.02	56.14	42.54	42.48	56.45	56.45	–	48.57	48.76	57.31	48.62	48.81	57.29	49.92	50.14	57.55
aniso 0.75/1.5 sigma0 + offset 0.5/-0.25 px, snr 35 dB	–/0.0	49.54	52.05	44.30	44.23	52.58	52.58	–	50.82	50.86	53.05	–	–	–	51.58	51.61	52.91
Pavia Centre ×4																	
nominal	0.00/0.0	49.63	49.37	50.63	50.63	50.40	50.40	50.41	50.45	50.45	50.45	50.42	50.42	50.42	50.43	50.43	50.43
sigma x0.5	0.00/0.0	43.13	49.09	32.71	50.18	50.19	49.84	50.26	40.48	50.31	50.31	41.38	50.25	50.25	40.82	50.28	50.28
sigma x0.75	0.00/0.0	47.78	49.35	40.85	50.57	50.38	50.33	50.41	46.83	50.45	50.45	47.13	50.41	50.41	46.89	50.42	50.42
sigma x1.25	0.00/0.0	48.94	49.35	45.37	50.62	50.38	50.36	50.40	48.83	50.43	50.43	48.89	50.40	50.40	48.71	50.42	50.42
sigma x1.5	0.00/0.0	48.16	49.33	42.54	50.59	50.36	50.32	50.38	47.33	50.42	50.42	47.49	50.38	50.38	47.31	50.41	50.41
sigma x2	0.00/0.0	47.26	49.29	40.35	50.53	50.33	50.26	50.35	45.80	50.39	50.39	46.03	50.34	50.34	45.87	50.38	50.38
shift -12 nm	0.00/0.0	47.60	47.14	48.48	47.47	50.19	48.52	50.26	48.75	50.28	50.28	47.77	50.32	50.32	47.66	50.30	50.30
shift -8 nm	0.00/0.0	48.63	48.34	49.64	48.83	50.30	49.42	50.35	49.62	50.38	50.38	49.07	50.38	50.38	48.97	50.37	50.37
shift -4 nm	0.00/0.0	49.35	49.07	50.39	50.13	50.37	50.11	50.40	50.22	50.43	50.43	50.05	50.41	50.41	49.98	50.41	50.41
shift +4 nm	0.00/0.0	49.43	49.19	50.40	50.31	50.40	50.18	50.41	50.26	50.45	50.45	50.04	50.41	50.41	50.00	50.44	50.44
shift +8 nm	0.00/0.0	48.86	48.56	49.80	49.28	50.37	49.51	50.38	49.73	50.42	50.42	49.12	50.39	50.39	49.11	50.42	50.42
shift +12 nm	0.00/0.0	48.06	47.54	48.96	48.12	50.32	48.59	50.31	48.98	50.37	50.37	47.98	50.35	50.35	48.03	50.39	50.39
snr 40 dB	0.00/0.0	47.95	48.12	46.91	46.91	48.71	48.71	48.76	48.87	48.87	48.87	48.91	48.91	48.91	48.77	48.77	48.77
snr 35 dB	0.00/0.0	45.68	46.29	43.43	43.43	46.75	46.75	46.86	47.10	47.10	47.10	47.18	47.18	47.18	46.90	46.90	46.90
snr 30 dB	0.00/0.0	42.10	43.19	39.10	39.10	43.72	43.72	43.91	44.30	44.30	44.30	44.44	44.44	44.44	43.99	43.99	43.99
sigma x1.5, shift +8 nm, snr 35 dB	0.00/0.0	44.82	45.84	40.17	42.43	46.72	46.36	46.85	45.29	47.07	47.07	45.34	47.14	47.14	45.21	46.88	46.88
sigma x0.75, shift -8 nm, snr 40 dB	0.00/0.0	46.28	47.20	40.03	46.57	48.64	48.04	48.71	45.93	48.83	48.83	46.06	48.88	48.88	46.03	48.74	48.74
sigma x2, shift +12 nm, snr 30 dB	0.00/0.0	41.35	42.63	36.67	37.95	43.69	43.29	43.91	42.48	44.27	44.27	42.52	44.39	44.39	42.33	43.97	43.97
aniso blur 1.5/0.75 sigma0, 30 deg	–/0.5	47.68	49.34	40.51	40.60	50.00	50.00	48.70	46.71	46.86	50.43	46.99	47.12	50.37	46.73	46.87	50.36
aniso blur 1.25/0.5 sigma0, 0 deg	–/1.0	46.18	49.30	38.16	40.65	49.88	49.68	47.84	44.29	45.08	50.39	44.82	45.39	50.22	44.54	45.14	50.32
disk blur r = 1.25 sigma0	–/1.0	46.80	49.31	38.93	45.85	50.14	49.99	49.39	45.35	48.38	50.42	45.79	48.54	50.29	45.46	48.27	50.32
MSI misregistered 0.5 px	0.03/2.2	31.75	31.18	32.40	32.43	31.35	31.37	31.65	31.94	31.97	–	31.93	31.97	–	31.96	32.00	–
MSI misregistered 1 px	0.85/5.0	26.31	25.69	27.04	26.93	25.81	25.83	26.08	26.58	26.48	–	26.47	26.37	–	26.54	26.42	–
aniso blur, MSI 0.5 px, snr 35 dB	–/1.8	31.53	31.11	31.55	31.57	31.34	31.35	31.53	31.75	31.76	–	31.68	31.70	–	31.74	31.76	–
HSI grid offset 0.5/0.5 px	–/2.2	45.99	49.33	40.01	39.96	49.40	49.24	–	44.71	44.70	50.45	45.11	45.04	50.43	44.91	44.83	50.45
HSI grid offset 1/-1 px	–/2.5	41.86	49.04	34.54	34.41	47.93	47.76	–	39.91	40.12	50.35	40.52	40.78	50.30	40.34	40.58	50.38
aniso 0.75/1.5 sigma0 + offset 0.5/-0.25 px, snr 35 dB	–/0.2	43.65	46.25	37.88	37.82	46.04	46.06	–	43.19	43.24	47.07	43.72	43.77	47.12	43.57	43.63	46.87

Table S23. Where the margin over the retrained networks comes from (4 h on one T4, single pass). The two strongest baselines are retrained with their authors' own recipe and with ExactFuse's; ExactFuse is reduced to its prior alone, to one stage, to three stages without the data step, and trained with the baselines' recipe. Δ: PSNR against the full ExactFuse of the same dataset; it.: training iterations completed.

	Chikusei ×4				Pavia Centre ×4			
Model and training	PSNR	Δ	SAM	it.	PSNR	Δ	SAM	it.
MIMFormer								
pipeline recipe (batch 8, AdamW, no decay, cosine)	56.96	-1.10	0.801	100k	49.63	-1.00	1.982	97k
authors' recipe (batch 16, Adam, step decay)	56.00	-2.05	0.874	68k	49.32	-1.30	2.022	68k
ExactFuse's recipe (lr 3×10^{-4}, decay, clipping, SAM+SSIM)	57.52	-0.53	0.760	81k	50.23	-0.40	1.954	84k
PSRT								
pipeline recipe (batch 8)	53.99	-4.06	1.057	64k	47.87	-2.75	2.212	65k
authors' recipe (batch 32, AdamW decay, clipping, constant lr)	49.51	-8.54	1.551	17k	45.44	-5.18	2.597	17k
ExactFuse								
prior alone: one U-Net, no data step	57.67	-0.38	0.747	113k	50.39	-0.23	1.934	115k
one stage with data step	57.75	-0.31	0.743	97k	50.60	-0.02	1.909	109k
three stages, no data step	57.65	-0.41	0.752	41k	50.46	-0.17	1.928	42k
baselines' recipe (lr 10^{-4}, no decay, no clipping, L1)	57.67	-0.38	0.749	32k	50.07	-0.55	1.921	35k
full model	58.05	–	0.724	30k	50.63	–	1.892	34k

Table S24. Complexity for one 256 × 256 Chikusei image (128 bands, ×4) on one Tesla T4.

Method	Params (M)	GFLOPs	Time (ms)	Memory (GB)
ExactFuse	10.45	412	221	0.54
Base model	10.13	296	181	0.46
SSR-NET	0.44	58	11	0.11
PSRT	0.30	43	122	0.42
DCT	8.46	2640	2255	1.10
MIMFormer	16.46	224	66	0.46
DHIF-Net	48.67	14930	2807	2.09
SMGU-Net	9.50	2286	511	0.98
CLSNet	4.17	142	170	0.36
OTPNet	2.51	328	155	0.34
MIMO-SST	5.32	81	29	0.33
CaFormer	7.83	222	171	0.57